Amazon Sidewalk Development Kit

Quick Start Guide

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Keyword

Amazon Sidewalk, Development Kit

Brief

This document provides a quick start guide for Amazon Sidewalk development kit.

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Revision History

Version	Revisions
V1.0.0	Initial Release
V1.0.1	Modified ML3238S pin assignment; added BLE/FSK/LoRa switching function to Sidewalk Demo.
V1.0.2	Added the Sidewalk SDK Overview section.

Contents

List of Figures

1 Overview

Amazon Sidewalk is a secure community network protocol for the Internet of Things (IoT) launched by Amazon. For detailed information, refer to the official website: https://sidewalk.amazon/

The protocol supports three wireless transmission methods:

- Amazon Sidewalk SubG-FSK, referred to as SubG-FSK
- Amazon Sidewalk SubG-CSS (based on LoRa® technology), referred to as SubG-CSS
- Bluetooth® Low Energy, referred to as BLE

The Amazon Sidewalk development kit is a comprehensive platform designed for Amazon Sidewalk applications. It comprises the Telink-developed AIOT-DK1 general-purpose development motherboard and ML3238S dedicated daughterboard. Supporting BLE, SubG-FSK, and SubG-CSS wireless communication protocols, the kit integrates functions including programming and debugging, functional development and demonstration, RF testing, and power consumption measurement, enabling rapid device prototyping and performance validation.

Figure 1.1: Amazon Sidewalk Development Kit

Related Documents & Tools

- AIOT-DK1 User Guide
- Sidewalk SDK User Guide
- Telink development tools:
  - Telink VS Code Extension User Guide
  - Burning and Debugging Tool (BDT)
  - Telink IoT Studio User Guide

2 Hardware Components

This section briefly introduces the four core hardware components of the kit: the AIOT-DK1 development motherboard, the ML3238S development daughterboard, the KGM100XB wireless module, and the on-board programmer. It outlines their functions, specifications, compatibility, and usage instructions to help users quickly set up the development environment.

2.1 AIOT-DK1 General Development Motherboard

The AIOT-DK1 is a multifunctional general development motherboard designed by Telink. It is used in conjunction with various Telink modules to perform functional development, verification, and demonstrations, serving as the core hardware platform for the entire kit. For the complete AIOT-DK1 hardware manual, pin definitions, circuit schematics, and usage specifications, please refer to the official documentation: AIOT-DK1_User_Manual.pdf.

Core Functions and Features:

- 3 types of on-board sensors, multiple LEDs, multiple buttons, and hardware reset;
- 3 on-board MikroBUS-compatible interfaces for flexible functionality expansion;
- On-board burning and debugging module supporting direct firmware burning and online debugging;
- Equipped with 3 USB Type-C ports for power supply, burning, and debugging;
- Supports module power consumption testing.

2.2 ML3238S Development Daughterboard

The ML3238S is a dedicated development daughterboard designed specifically for Amazon Sidewalk. It should be used in conjunction with the AIOT-DK1 development motherboard and cannot operate independently; it serves as the key platform for implementing Amazon Sidewalk-related functions.

The core hardware configuration is as follows:

- On-board KGM100XB wireless communication module, providing wireless communication support for Sidewalk functionality;
- Integrated 2.4 GHz & 900 MHz dual-band chip antenna, ensuring stable wireless signal transmission;
- Equipped with an RF coaxial connector with a switch, supporting external antenna expansion and RF performance testing.

Subsequent sections provide the complete schematic of the ML3238S, hardware connection details with the AIOT-DK1 motherboard, pin assignments, and debugging considerations.

2.3 KGM100XB Wireless Communication Module

The KGM100XB is a wireless module designed by Quectel specifically for Amazon Sidewalk. It supports dual-link communication featuring BLE, SubG-FSK, and SubG-CSS, making it ideal for low-power, long-range devices in IoT and smart home applications.

Core chip components:

- Telink TL323x SoC: The main controller, responsible for BLE communication and MCU functions.
- Semtech SX126x Transceiver: A high-performance LoRa/FSK transceiver responsible for long-range, low-power communication.

2.4 On-board Programmer

The on-board programmer is a dedicated burning and debugging unit integrated onto the AIOT-DK1 motherboard, enabling firmware burning and hardware debugging without the need for an external debugger.

Key Advantages:

- Integrated on-board design eliminates the need to purchase an external debugger, simplifying the development process.
- Supports three debugging modes: Single-Wire (Swire), SDP, and JTAG.
- Equipped with status indicators for intuitive display of burning and debugging status.
- Supports one-click program burning, Flash erasure, chip activation, and memory read/write operations.

This kit should be used in conjunction with the Burning and Debugging Tool (BDT).

- Telink official burning and debugging software for Windows, Linux, and macOS.
- Supports firmware download, Flash erasure, security encryption and decryption, variable debugging, and log viewing.
- Official tool download link: Telink | Development Tools
- Detailed user guide (including GUI and command line): Burning and Debugging Tool (Windows)

3 ML3238S-C1T407A67_V1.x Schematic Design

The ML3238S schematic centers on U1 (module interface) and U2 (KGM100XB), connecting directly to the J2 connector on the AIOT-DK1 motherboard. It supports pin multiplexing, RF switching, and a dual-antenna design.

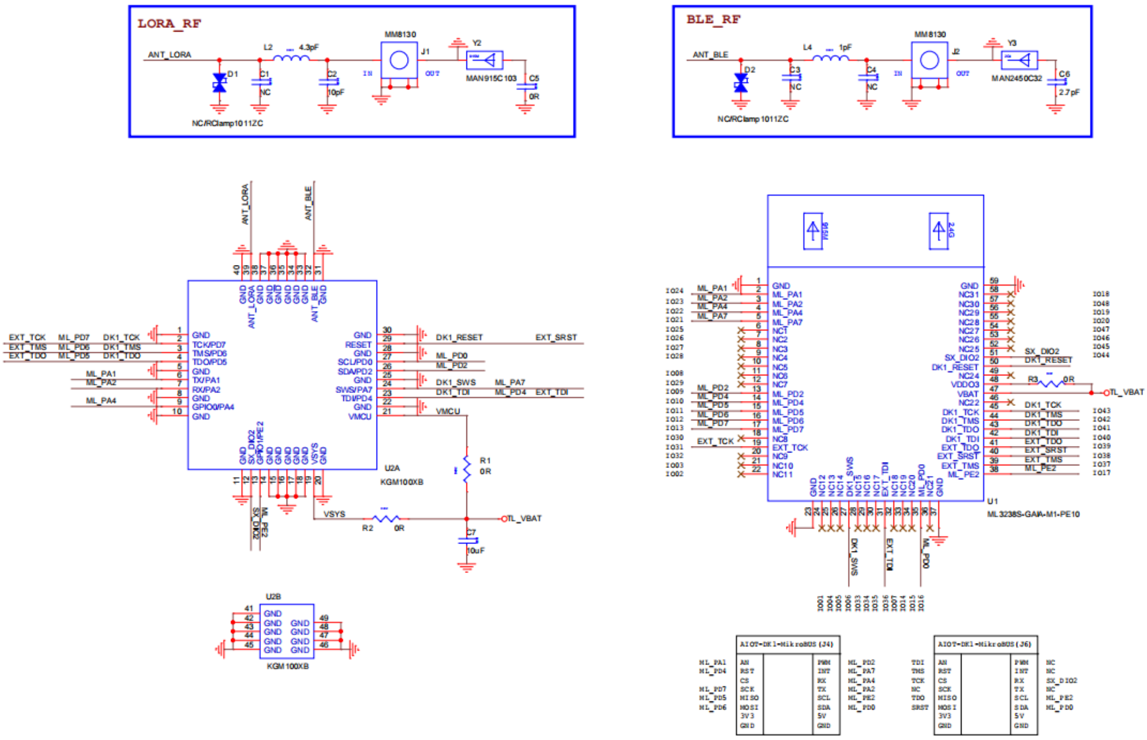

Figure 3.1: ML3238S-V1.x Schematic

The IO connection with the AIOT-DK1 is shown in the figure below.

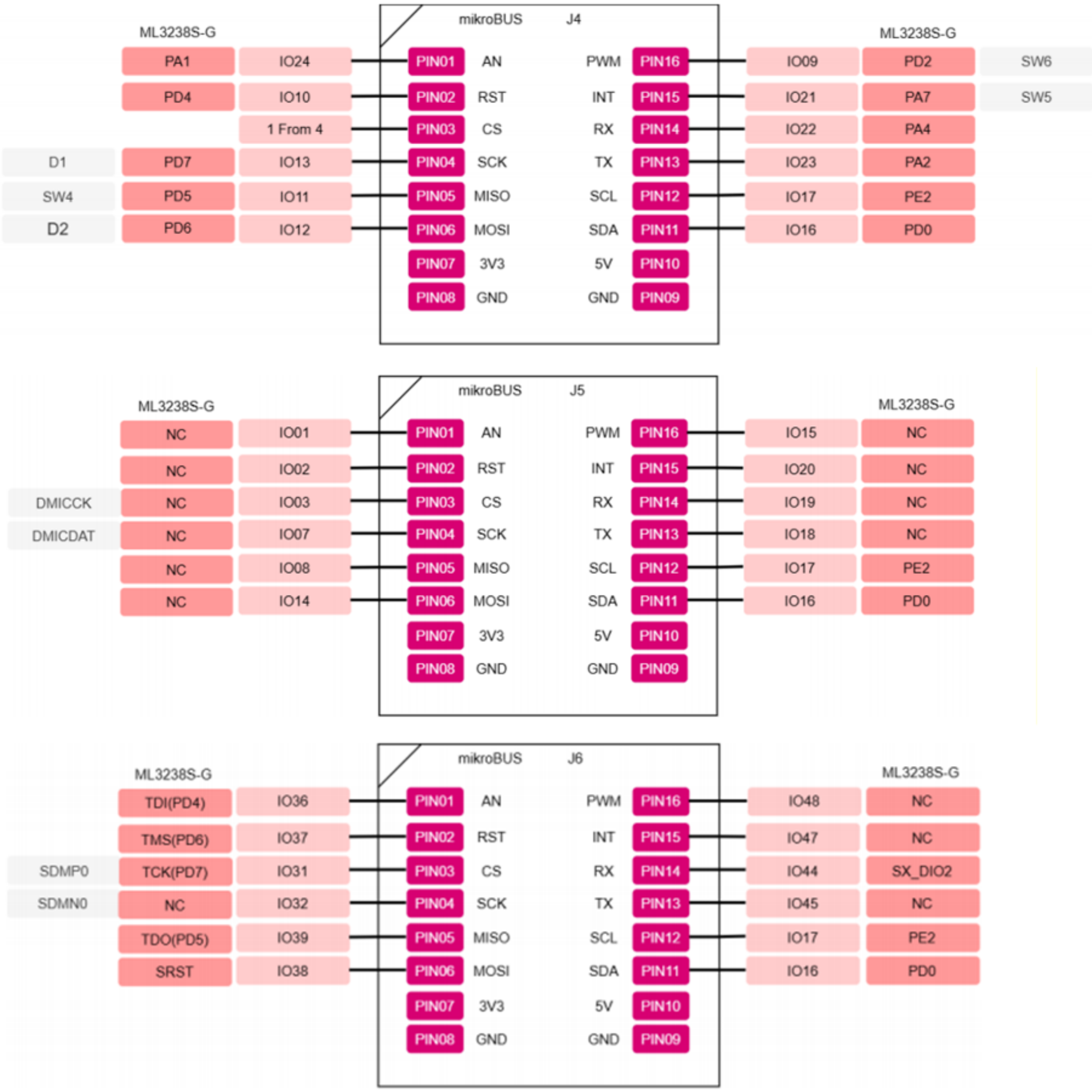

Figure 3.2: J4/J5/J6 Connection with ML3238S

3.1 U1 Pin Assignment and Multiplexing Description

U1 serves as the external interface of the ML3238S module and corresponds to AIOT-DK1's J2 connector. To enhance flexibility, some GPIOs support function multiplexing and can be configured by the user according to application scenarios.

3.1.1 JTAG & All-Function GPIO Multiplexing

PD4/PD5/PD6/PD7 support both All-Function GPIO and JTAG debugging modes:

- ML_PD4/PD5/PD6/PD7: Connect to MikroBUS (J4) and can be configured as standard GPIOs for peripheral expansion.
- DK1_TCK/TMS/TDO/TDI: Connect to the AIOT-DK1 on-board programmer for on-board JTAG debugging.
- EXT_TCK/TMS/TDO/TDI: Connect to MikroBUS (J6) for external debugging using standard debuggers such as J-Link.
- SW8 needs to be switched to the left side.

Note:

> - For All-Function GPIO definitions, please refer to the TL323x chip datasheet (contact Telink sales or FAE for this document).

3.1.2 SWS & GPIO Multiplexing

The chip supports a single-wire communication interface (Single Wire Slave). The SWM/SWS protocol is a Telink single-wire master/slave protocol with a maximum data rate of 2 Mbps, enabling burning and debugging via the on-board programmer:

- DK1_SWS: Connects to the AIOT-DK1 on-board programmer for firmware burning and online debugging.
- ML_PA7: The SWS pin is multiplexed as a standard GPIO and connected to the MikroBUS (J4), allowing it to be used as a general-purpose I/O.

3.1.3 Reset & SRST Multiplexing

- DK1_RESET: Connects to the AIOT-DK1 reset button (SW2) and supports hardware reset.
- EXT_SRST: Connects to the JTAG standard interface SRST, connecting with the J-Link reset pin for tool-controlled reset.

3.2 U2 (KGM100XB) Power Supply Design

U2 is the KGM100XB core module, which contains two chips internally. The module's power supply configuration is as follows:

- VMCU: Power supply for TL323x (MCU/BLE)
- VSYS: Power supply for SX126x (SubG-FSK/SubG-CSS)

Note:

> - VMCU and VSYS are shorted together and are powered uniformly by VBAT (3.3V by default). The power supply circuit includes a reserved test resistor that can be disconnected to enable independent current testing for TL323x and SX1262.

3.3 RF and Antenna Design

RF Switch Connector (MM8130): The Murata MM8130 RF coaxial connector (SWF series, SMD package) is used:

- Features: 50Ω impedance, supports 6 GHz bandwidth, built-in test switch.
- Function: Allows disconnection of the on-board antenna and connection to external instruments for RF debugging, calibration, and testing.

Chip Antenna

- 2.4 GHz Chip Antenna: Dedicated to BLE communication.
- 900 MHz Chip Antenna: Dedicated to SubG-FSK/SubG-CSS long-range communication.
- Compact SMD Design: Facilitates product integration and development verification.

4 ML3238S-C1T407A67_V2.x Schematic Design

The ML3238S schematic centers on U1 (module interface) and U2 (KGM100XB), connecting directly to the J2 connector on the AIOT-DK1 motherboard. It supports pin multiplexing, RF switching, and a dual-antenna design.

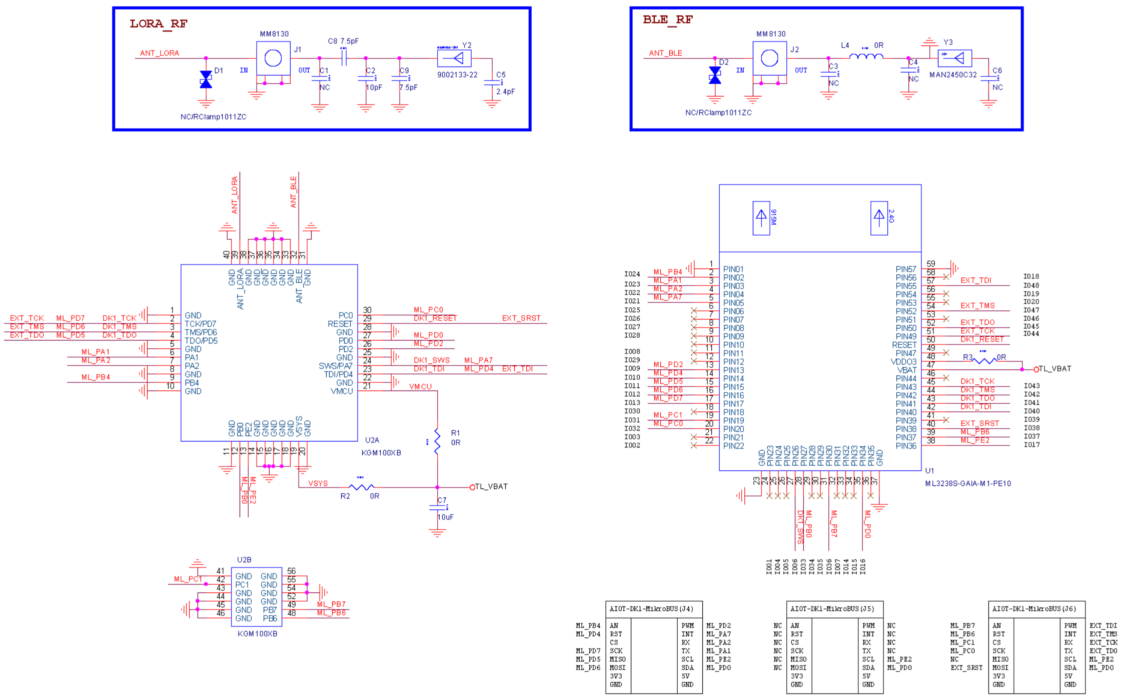

Figure 4.1: ML3238S Schematic

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5 AIOT-DK1 Features Overview

The latest AIOT-DK1 Hardware Manual: AIOT-DK1_User_Manual.pdf

The key features of **ML3238S + AIOT-DK1** are summarized as follows:

No.	Features	ML3238S-G	Instruction
1	Digital microphone	-	Toggle SW7 left
2	SDM JACK	-	Toggle SW8 down
3	Camera	-	Toggle SW17 left
4	Module reset - SW2	Y	-
5	Normal button - SW4	Y	-
6	Normal button - SW5	Y	-
7	Normal button - SW6	Y	-
8	LED - D1 (White)	Y	-
9	LED - D2 (Green)	Y	-
10	IR	-	-
11	USB Full-speed	-	-
12	USB High-speed	-	-
13	Pressure sensor	Y	-
14	Relative humidity and temperature sensor	Y	-
15	3D accelerometer, 3D gyroscope, 3D magnetometer sensor	Y	-

5.1 Power Supply Methods

The AIOT-DK1 motherboard supports three USB Type-C power supply methods. For detailed configuration and jumper settings, refer to Section 2.1 Power Supplies.

5.2 Current Measurement

The AIOT-DK1 motherboard supports precise measurement of module power consumption and sensor current. For testing methods, jumper configuration, and wiring procedures, refer to Section 2.3 Current Measurement.

5.3 Debug Interface

The AIOT-DK1 supports three debugging methods: single-wire, JTAG, and SDP. For detailed configuration, refer to Section 2.2 Debug Interface.

Firmware Burning and Debugging Tool: BDT Tool

- Official BDT download link: Telink | Development Tools

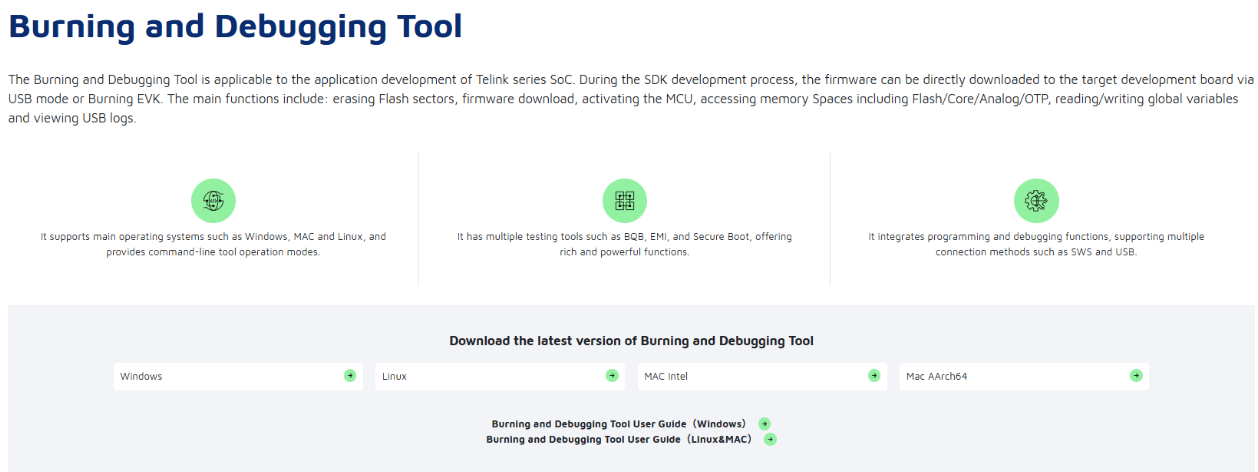

Figure 5.1: BDT Download

- Refer to the BDT user guide: Burning and Debugging Tool (Windows)

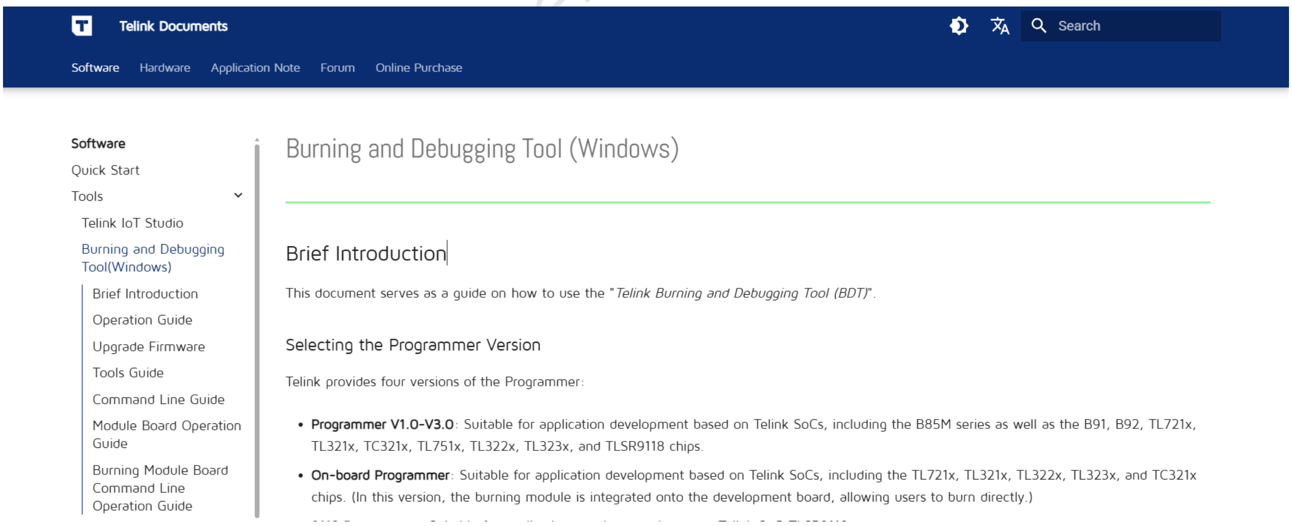

Figure 5.2: BDT User Guide

- Operating Notes
  - Drivers should be installed for first-time use (one-click installation available in the tool interface).
  - If the chip is in sleep mode, use the `Activate` function to wake it up before burning.

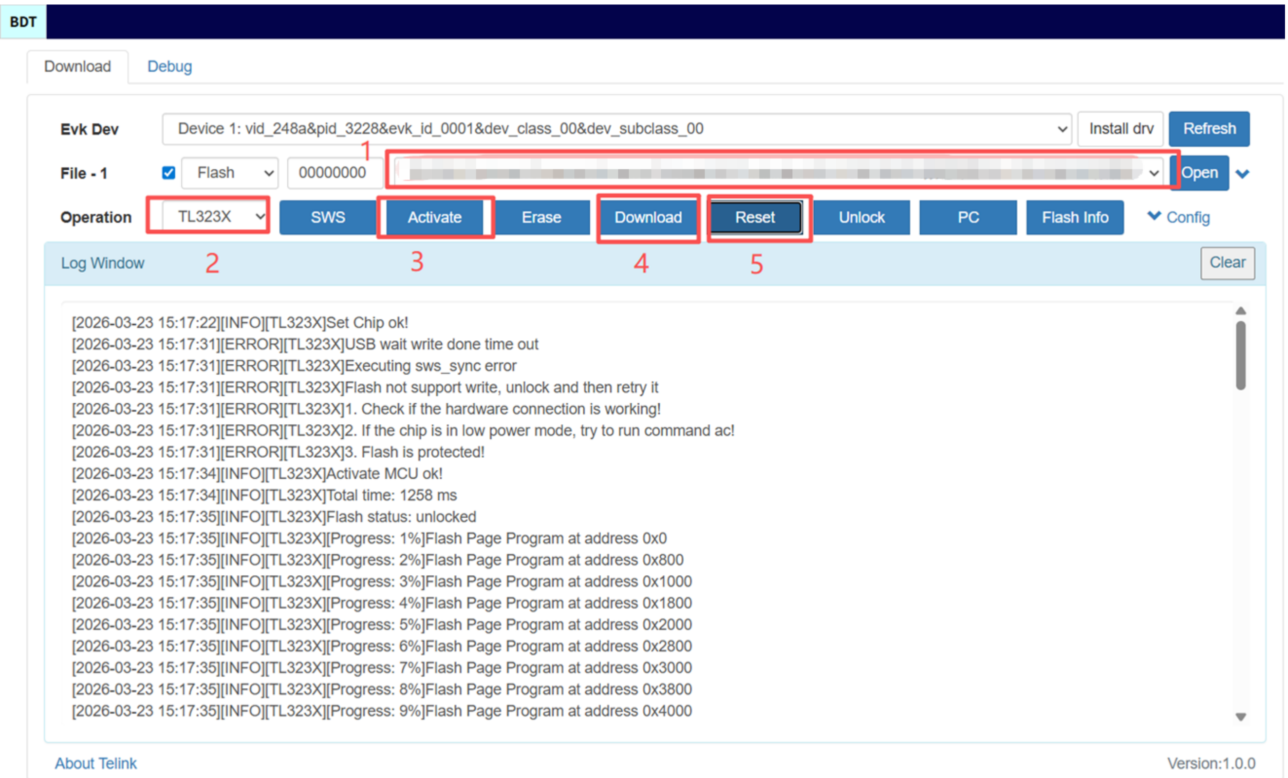

Figure 5.3: Operating Notes

6 Sidewalk SDK Overview

The Sidewalk SDK for the Telink TL323x SoC, targeting the AIOT-DK1 development kit with ML3238S module.

This section describes how to get started with the Sidewalk SDK.

- The first part focuses on setting up the AWS environment and allows you to run the demo (Sidewalk sensor monitoring).
  - For now, the sensor monitoring demo is only available as a binary, which can be downloaded at sidewalk_sensor_monitoring.bin.
  - Moreover, the MFG.bin file needs to be programmed into the board. The MFG generation method refers to Prepare the MFG blob chapter.
- The second part explains how to use the Amazon Sidewalk DUT. Since Sidewalk applications require provisioning data (a specially prepared binary blob), it is recommended to go through the first part even if you do not intend to use the sidewalk_sensor_monitoring demo.

6.1 Prerequisites

6.1.1 Hardware

There are four important components involved:

- AWS cloud infrastructure running a demo web app
  - Refer to the tutorial Build a Basic AWS Web Application, which uses AWS Amplify to build a full-stack web application.
- Sidewalk gateway
  - Echo 4 or other compatible Sidewalk gateways.
  - For detailed gateway models and their supported communication protocols (BLE, SubG-CSS, SubG-FSK), refer to List of Sidewalk Gateways
- Sidewalk end device
  - Demonstration with Telink Sidewalk Kit (AIOT-DK1 + ML3238S)
    - AIOT-DK1 motherboard: Refer to AIOT-DK1 User Manual, which supports multiple modules, integrates 3 sensors, 3 mikroBUS interfaces, and supports on-board burning.
    - ML3238S User Manual - To be updated
  - UART-USB converter for logs (optional)
  - Male-female Arduino jumper cables
- Open VPN capable Wi-Fi router

  As of March 2026, direct access is limited to the United States, Canada and Mexico. Users in other regions must connect to the server via a VPN.

6.1.2 Software

Install the required toolchains, programming, and debugging tools. Official Tools Download Link: Telink Development tools

- Telink VS Code Extension
  - Telink VS Code Extension is an all-in-one embedded development tool specifically built for Telink chips. Deeply integrated into the ecosystem of Visual Studio Code (VS Code) and its derivative AI editors, it is designed to provide developers with an efficient full-lifecycle development experience covering environment configuration, project creation, coding, compilation, debugging, and firmware burning.
  - Refer to Telink VS Code Extension User Guide.
- Burning and Debugging Tool (BDT)
  - The Burning and Debugging Tool is applicable to the application development of the Telink series SoC. During the SDK development process, the firmware can be directly downloaded to the target development board via USB mode or Telink Programmer. The main functions include: erasing Flash sectors, firmware download, activating the MCU, accessing memory spaces including Flash/Core/Analog/OTP, reading/writing global variables, and viewing USB logs.
  - Refer to Burning and Debugging Tool User Guide.
- Python 3.9+
- AWS CLI
- Git

6.2 Set Up the Sidewalk Gateway

Until March 2026, Amazon Sidewalk is available only in North America (the US, Canada, and Mexico). In order to develop Sidewalk solutions outside North America, the gateway device must be "tricked" into being USA-located. This can be done by VPN capable Wi-Fi router. "Gl.Inet GL-MT300N-V2 Router Mango Smart Mini" device with a Nord VPN account is proven to work.

Amazon suggests using open VPN protocol for this network purpose. Open VPN provided by Nord VPN has been used in this demo with success. No other VPN provider nor VPN protocol has been tested.

Amazon Sidewalk uses 902-928 MHz for sub-giga operation. In the EU, this frequency range collides with the LTE band, which is strictly protected by UKE (Office for Electronic Communication) in Poland and its equivalent government agencies in other European countries. In order to develop sub-giga capabilities of sidewalk (which are out of scope for this demo), a coaxial cable is recommended instead of using over-the-air transmission.

Note:

> - European availability is planned for 2026, but has not yet been announced. Messaging for EU (European Union) audiences should avoid implying current coverage. Coverage maps are available for this region: Sidewalk Coverage.

A shielded room/cage should be used for outside-US operation of sub-giga modes (FSK, LoRA). In order to reduce development cost, an RF shielded cage can be avoided. The Echo 4 device can be converted for cable operation with u.lf -> SMA "pigtail". Using a 40~60dB attenuator is recommended to avoid damage to the radios. This approach, however, requires some basic RF and hardware handling skills and is not officially recommended.

Refer to the Amazon documentation in order to set up the Echo 4 or other compatible sidewalk gateway. It is a good starting point to follow this link: Setup an Amazon Echo as a Sidewalk Gateway in a Non-US Location.

6.3 Set Up the AWS Cloud

Go through the following steps to deploy the Amazon reference app and provision your device.

Your outcome shall be a working **web application** and **MFG binary blob** for commissioning.

6.3.1 Clone the AWS Sample App

Clone the web application's repository:

```
git clone https://github.com/aws-samples/aws-iot-core-for-amazon-sidewalk-sample-app.git
cd aws-iot-core-for-amazon-sidewalk-sample-app
```

6.3.2 Set Up Python Environment

Setting a virtual Python environment is the preferred way of proceeding in this document.

Install pyenv, referring to Simple Python Version Management: pyenv according to your system (Linux or Windows).

Download preferred Python version:

```
pyenv install 3.9.18
```

Set up and activate the virtual environment for the AWS repository (still in the AWS repository folder).

```
pyenv virtualenv 3.9.18 sidewalk-aws-39
pyenv local sidewalk-aws-39
```

Check whether it is working by verifying the Python version.

```
python -V
```

It should show 3.9.18.

Use pip (under a virtual environment) to install all needed packages.

```
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r requirements.txt
python -m pip install pyjwt -t ./ApplicationServerDeployment/lambda/
```

6.3.3 Set up AWS Account

Set up an AWS account (the account itself is free to create, and the demo does not require any paid features).

After having an AWS root account, set up an IAM "sub-account". This is described by Amazon, and it is not described in this document.

Store the account ID, which is needed for IAM user login.

Note:

> - The word "Console" used by Amazon refers to the web interface controlling AWS accounts. This is not a console in "normal" embedded understanding.

Generate access key for IAM user:

Log in to the root console and navigate to the IAM user's panel:

root console -> IAM (already invoked or use "search" bar) -> Access Management -> Users -> click on user's name

Use "Create access key" and store the resulting credentials pair:

Key ID should look like

```
AKIAIOSFODNN7EXAMPLE
```

and the secret key should look like:

```
wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY
```

Store these for further use.

6.3.4 Configure AWS CLI

AWS is a command-line tool to interact with Amazon servers. First, you need to configure it.

```
aws configure
```

Follow the prompts and provide credentials for your IAM user.

6.3.5 Set Up IAM Permissions

The IAM account needs privileges in order to deploy and run the app. Amazon provides a script that generates all* needed privileges in the form of JSON.

Run the Python script and store its output:

```
python ApplicationServerDeployment/policies/generate_policy.py
```

Go back to the AWS root user console and navigate to IAM -> Policies -> Create Policy (yellow button on top right). Switch to JSON and copy-paste the result from the script above. Give the policy a meaningful name like sidewalk_policy.

Add policy to your IAM user:

Navigate (still as a root user) to the User panel IAM -> Access Management -> Users -> (your IAM user name) -> Add Permission -> Attach policies directly -> search for your newly created policy.

During experiments, it turned out that policies stated by the script are not enough. Here comes a set of additional ones, which have to be added. You can do it by: IAM -> Access Management -> Users -> (your IAM user name) -> create inline policy -> JSON

bucket_tagging

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AllowS3BucketTagging",
            "Effect": "Allow",
            "Action": [
                "s3:PutBucketTagging",
                "s3:GetBucketTagging",
                "s3:DeleteBucketTagging"
            ],
            "Resource": "arn:aws:s3:::*"
        }
    ]
}
```

wireless_tagging

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AllowIoTWirelessTaggingDestination",
            "Effect": "Allow",
            "Action": [
```

```
                "iotwireless:TagResource",
                "iotwireless:UntagResource",
                "iotwireless:ListTagsForResource"
            ],
            "Resource": "arn:aws:iotwireless:us-east-1:488339498666:Destination/
            SensorAppDestination"
        }
    ]
}
```

wireless_full

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "IoTWirelessFullAccess",
            "Effect": "Allow",
            "Action": "iotwireless:*",
            "Resource": "*"
        }
    ]
}
```

Additionally, grant full access to AmazonDynamoDB:

(From user panel) -> Add Permissions -> Add Permissions -> Attach policies directly search for

```
AmazonDynamoDBFullAccess
```

6.3.6 Deploy the Sensor Monitoring Web App

Edit the config.yaml file:

```
*HARDWARE_PLATFORM* set to anything you like, NORDIC seems to be OK
*USERNAME* change to preferred user name for demo web app
*PASSWORD* set password for demo web app
```

Run deploy script:

```
python ApplicationServerDeployment/deploy_stack.py
```

The deployment takes several minutes. Upcoming errors (if any) are mostly related to missing privileges. It was tested OK against the commit ID:

```
cbf390429095b90db3cad36df106560270d32f38
```

The privileges and policies generated with the script, as presented in this document, were tested against this commit ID.

6.3.7 Ensure the Web App Is Up and Running

Open

```
aws-iot-core-for-amazon-sidewalk-sample-app/config.yaml
```

and look for:

```
    USERNAME: awsome_user
    PASSWORD: very_top_secret_password

    WEB_APP_URL: blah_blah.cloudfront.net
```

Use WEB_APP_URL and credentials. See the web app, but no devices yet.

6.4 Prepare the MFG Blob

The MFG blob contains device provisioning data (certificates and keys) that the Sidewalk stack needs at runtime.

6.4.1 Provision a Device in AWS

Run the provisioning script:

```
python EdgeDeviceProvisioning/provision_sidewalk_end_device.py
```

Note:

> - The MFG blob generated by this script is not directly compatible with the Telink demo. Follow the steps below to generate a compatible version.

6.4.2 Extract the Device Certificate

Sign in to the AWS website https://us-east-1.console.aws.amazon.com/iot/home?region=us-east-1#/wireless/sidewalk/devices, open AWS root user console, go to AWS IoT (use search bar if not visible) -> Manage -> LPWAN devices -> Sidewalk -> Devices

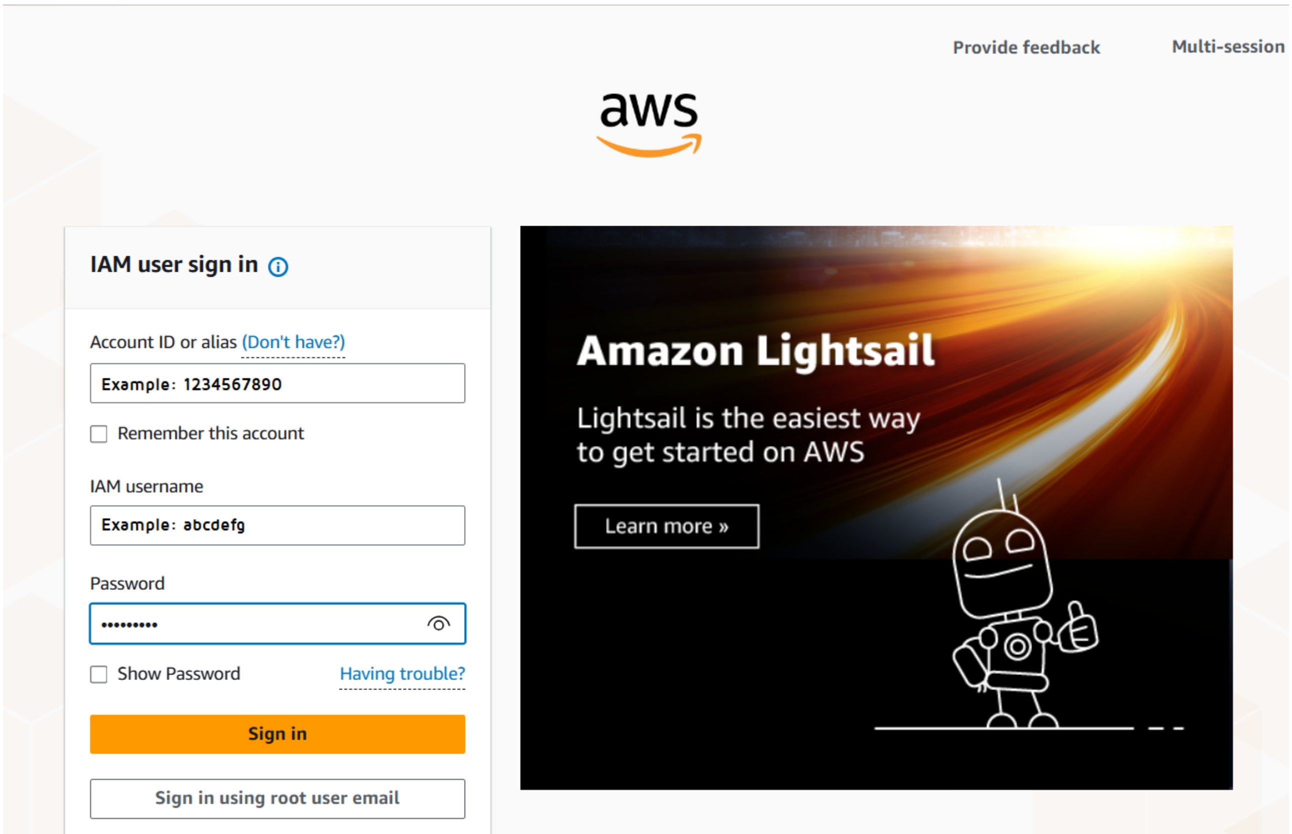

Figure 6.1: AWS Sign-In Page

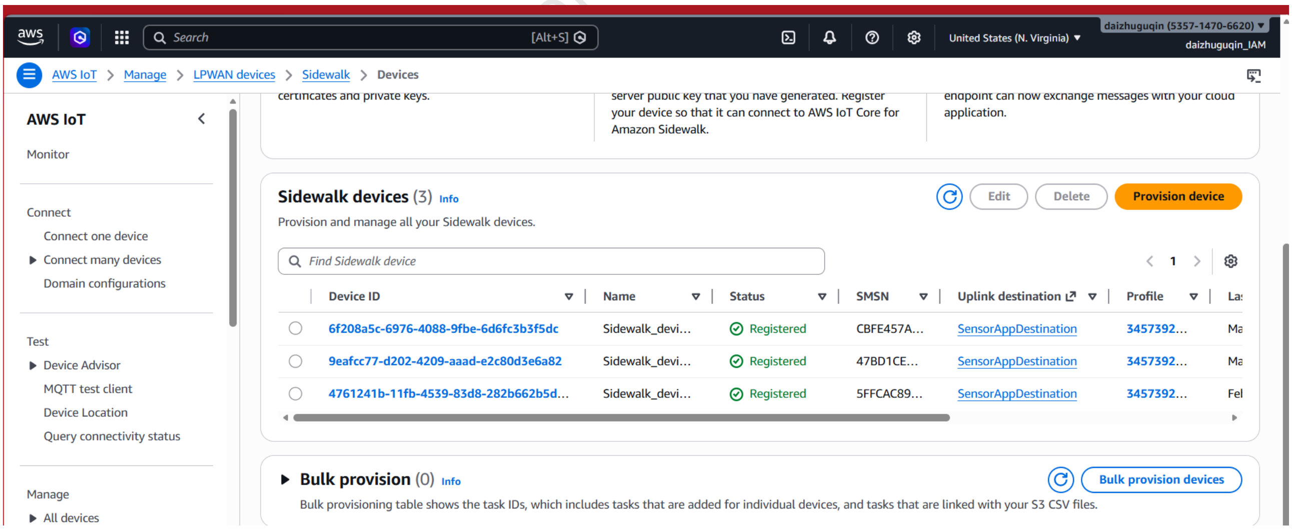

Figure 6.2: Device List Page

Detailed steps:

(1) Enter a custom device name.

(2) Select the profile: Sidewalk_profile

(3) Select the uplink destination: SensorAppDestination

(4) Click `Create` .

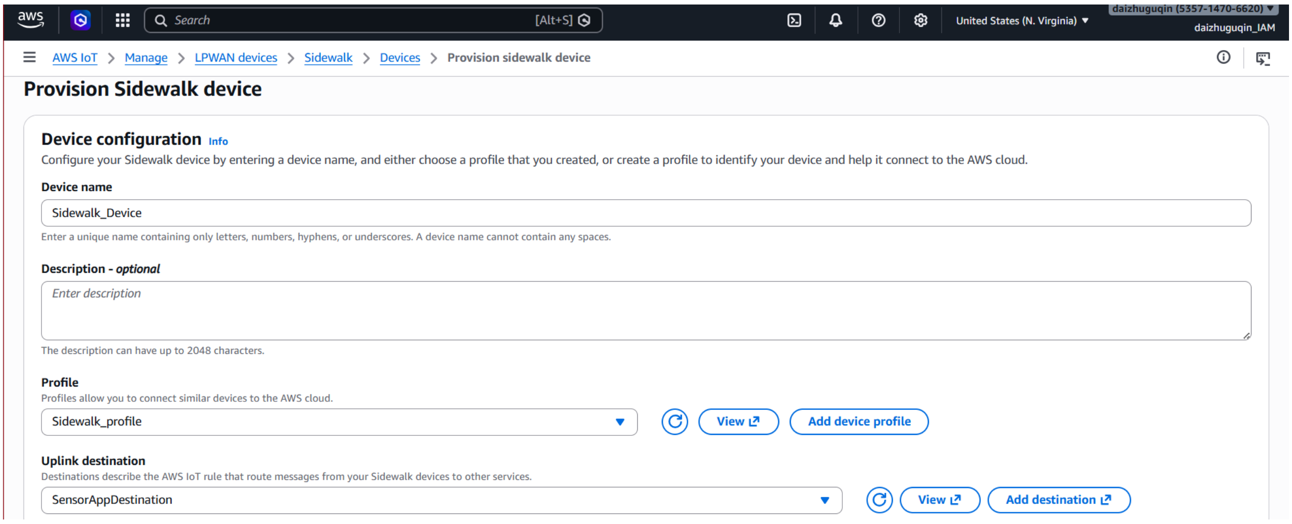

Figure 6.3: Device Provisioning Page

After the device is successfully created, you can view it in the `Sidewalk Devices` list.

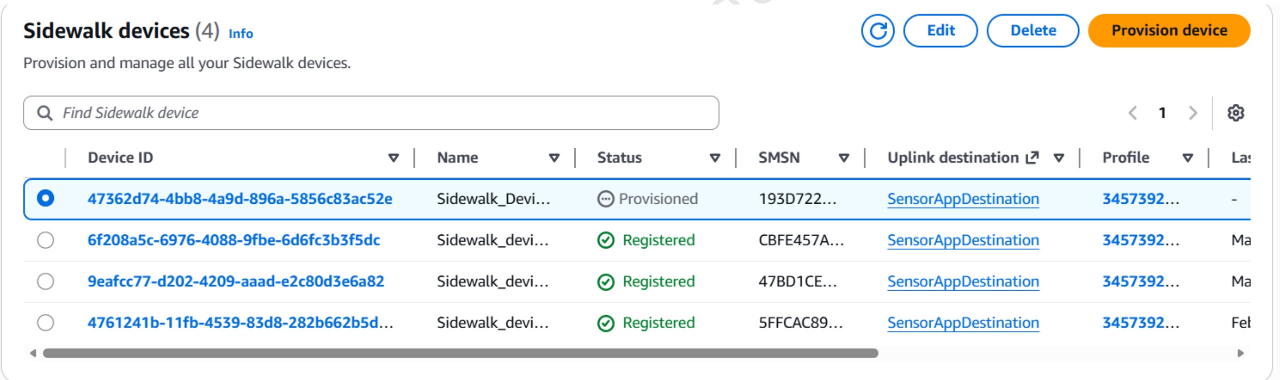

Figure 6.4: Sidewalk Devices Page

Your newly provisioned device should appear on the list. Click the device ID to open the device panel. Use Download device JSON file to download the device certificate, and save it for the next step.

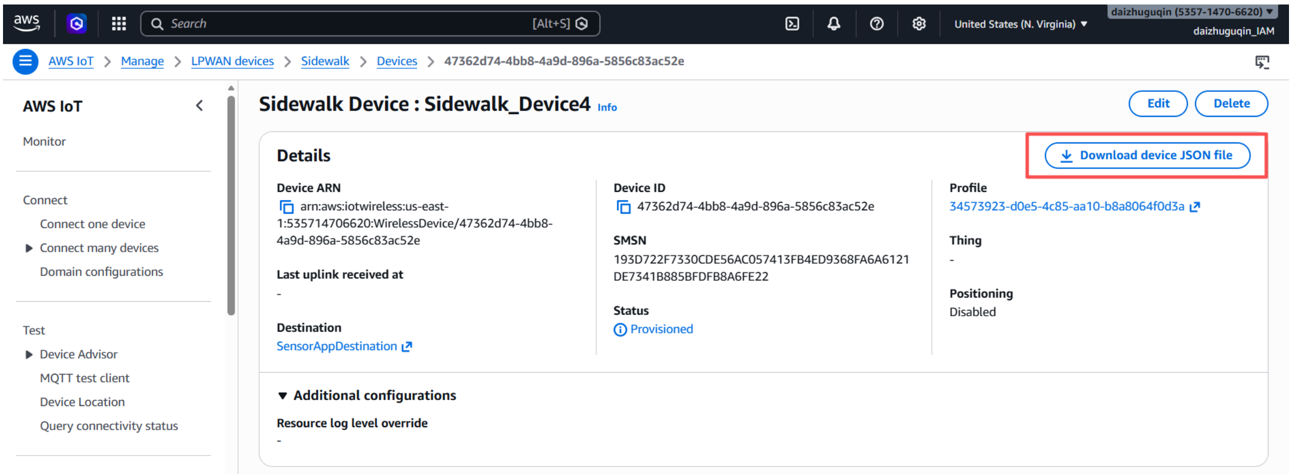

Figure 6.5: Download Device JSON File

6.4.3 Generate the MFG Binary

In Telink Sidewalk SDK navigate to:

```
tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/tools/provision/
```

Copy the certificate.json file to the current directory.

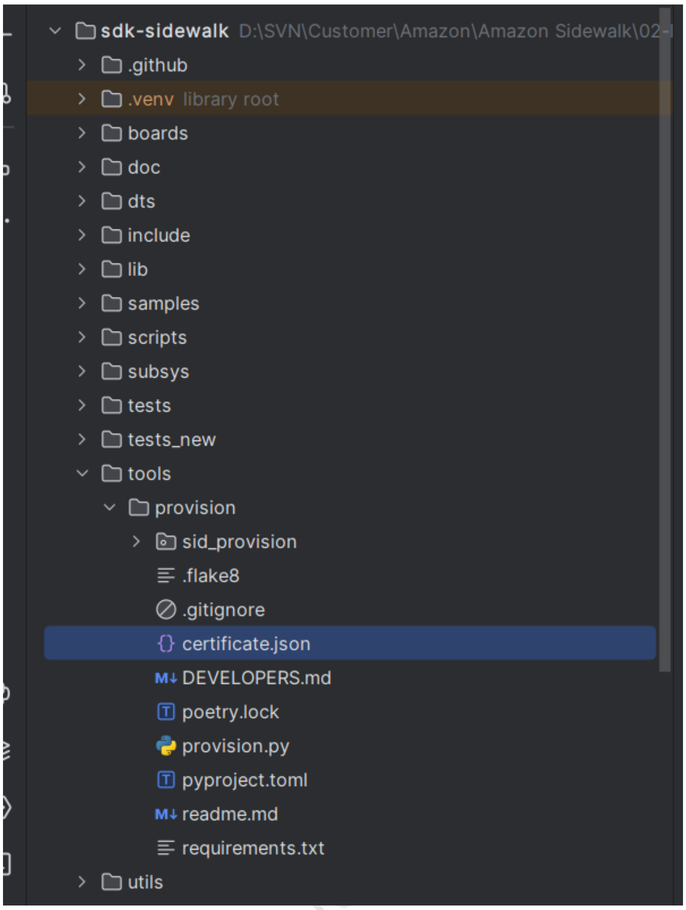

Figure 6.6: Current Directory

Use python script to generate MFG binary. Note that the mfg.bin file generated is identical regardless of the addr values used.

```
python3 provision.py nordic aws --output_bin mfg.bin \
--certificate_json certificate.json \
--addr 0x1F4000
```

The certificate.json is a file gathered in previous step.

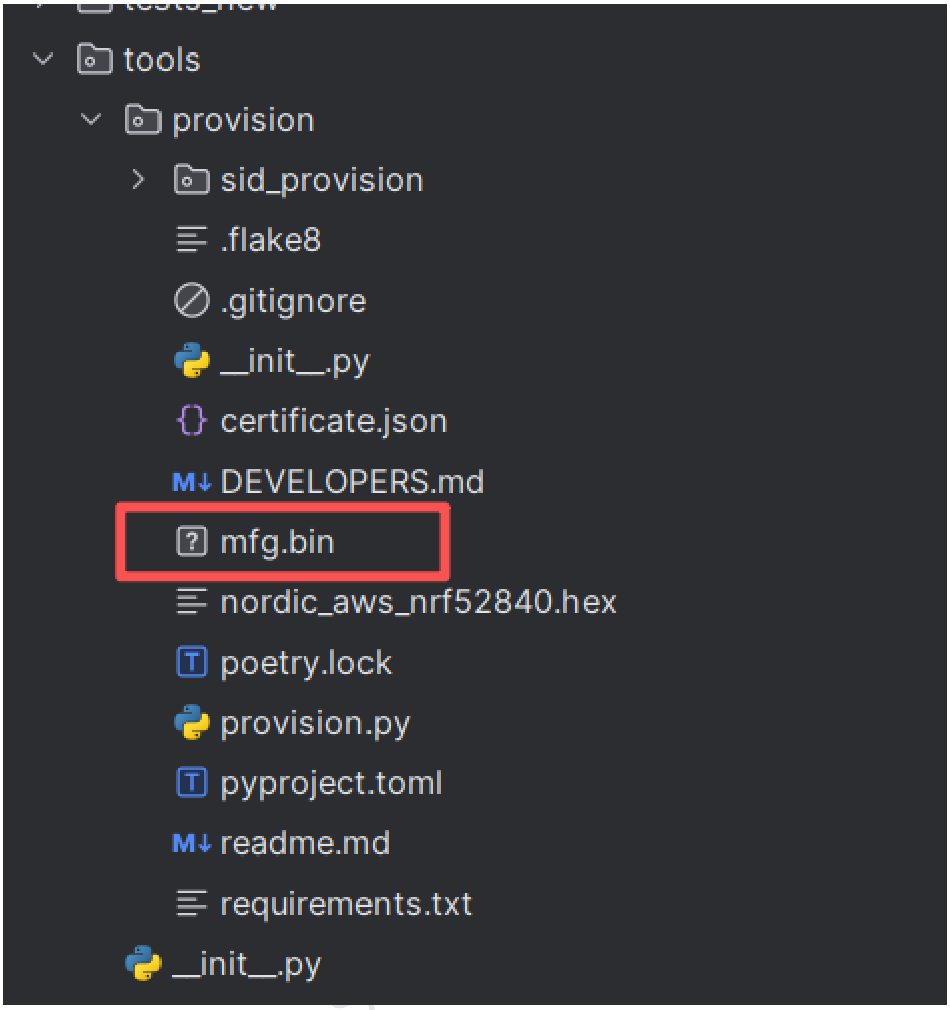

Figure 6.7: Generated MFG File

If the command line operation fails, you can also configure it via the "Edit Configuration" page as follows:

(1) Click **Edit Configuration...** in the top-right corner of PyCharm.

(2) Configure the Python interpreter: Select any available Python version (any version is acceptable)

(3) Specify the PythonScript path: Select the path to provision.py

(4) Paste this line directly into **Script Parameters** (copy the entire segment below):

```
nordic aws --output_bin mfg.bin --certificate_json certificate.json --addr 0x1F4000
```

(5) Click `OK` to save.

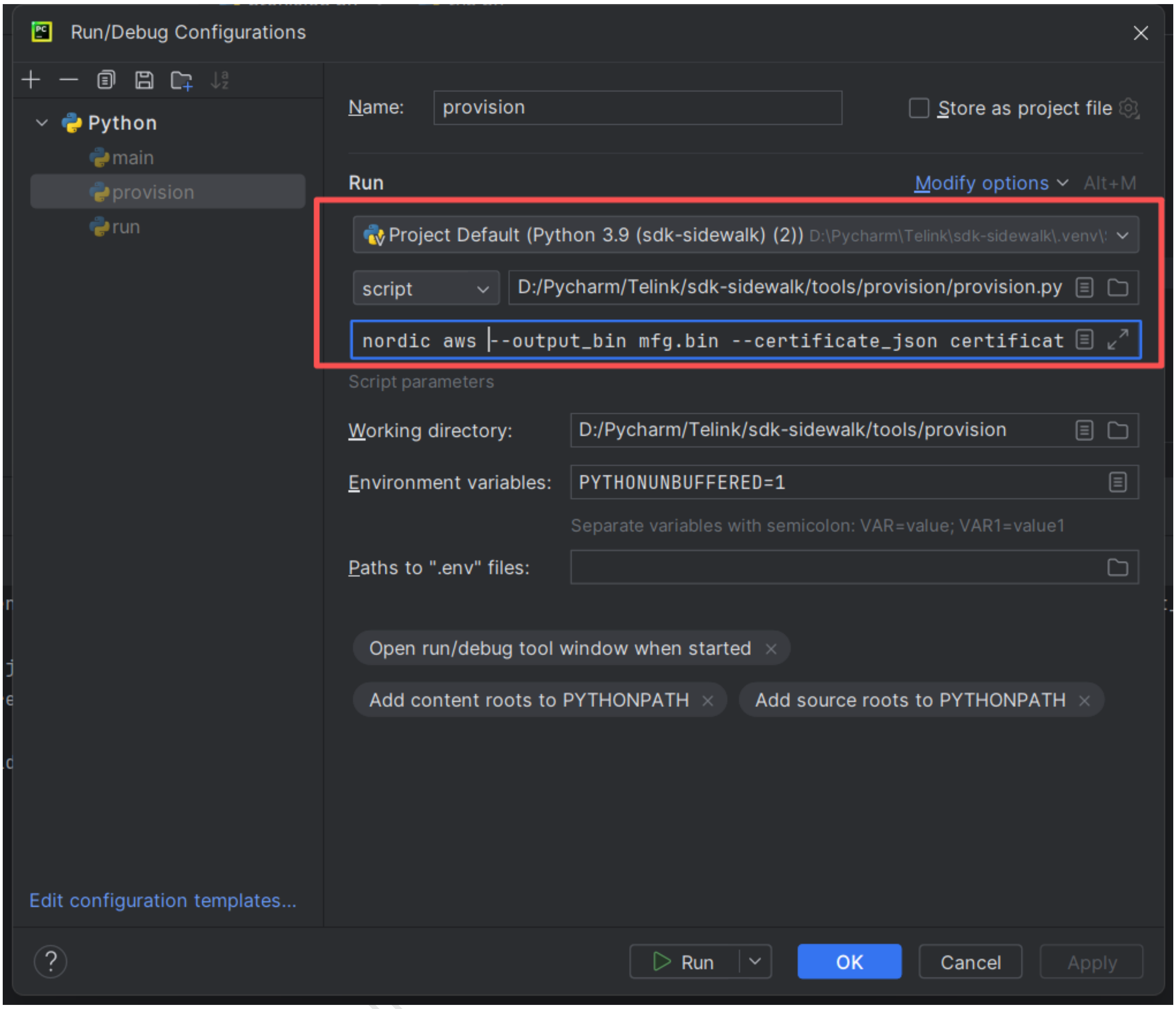

Figure 6.8: Edit Configuration Page

(6) Run the provision.py to generate the mfg.bin file.

```
D:\Pycharm\Telink\sdk-sidewalk\.venv\Scripts\python.exe D:\Pycharm\Telink\sdk-sidewalk\tools\provision\provision.py nordic aws --output_bin mfg.bin --certificate_json
 certificate.json --addr 0x1F4000
deviceTypeId found in certificate.json
Using chip config : (nrf52840 address: 0x1f4000)
Generated mfg.bin
Generated D:\Pycharm\Telink\sdk-sidewalk\tools\provision\nordic_aws_nrf52840.hex

Process finished with exit code 0
```

Figure 6.9: Run provision.py

Note:

- For more information on generating MFG files, please refer to Provision your Sidewalk endpoint.

6.5 Build the Firmware

GitHub Repository: https://github.com/telink-semi/tl_amazon_sidewalk

Execute the following commands to clone the repository and switch to the specified branch:

```
$ git clone https://github.com/telink-semi/tl_amazon_sidewalk
$ git checkout sidewalk_amazon_demo_and_LCD
```

cd into the project folder and start VS Code

```
$ cd tl_amazon_sidewalk
$ vscode
```

Open Telink extension in VS Code and find "Project Outline" sidebar. Use "convert" button if projects are not visible.

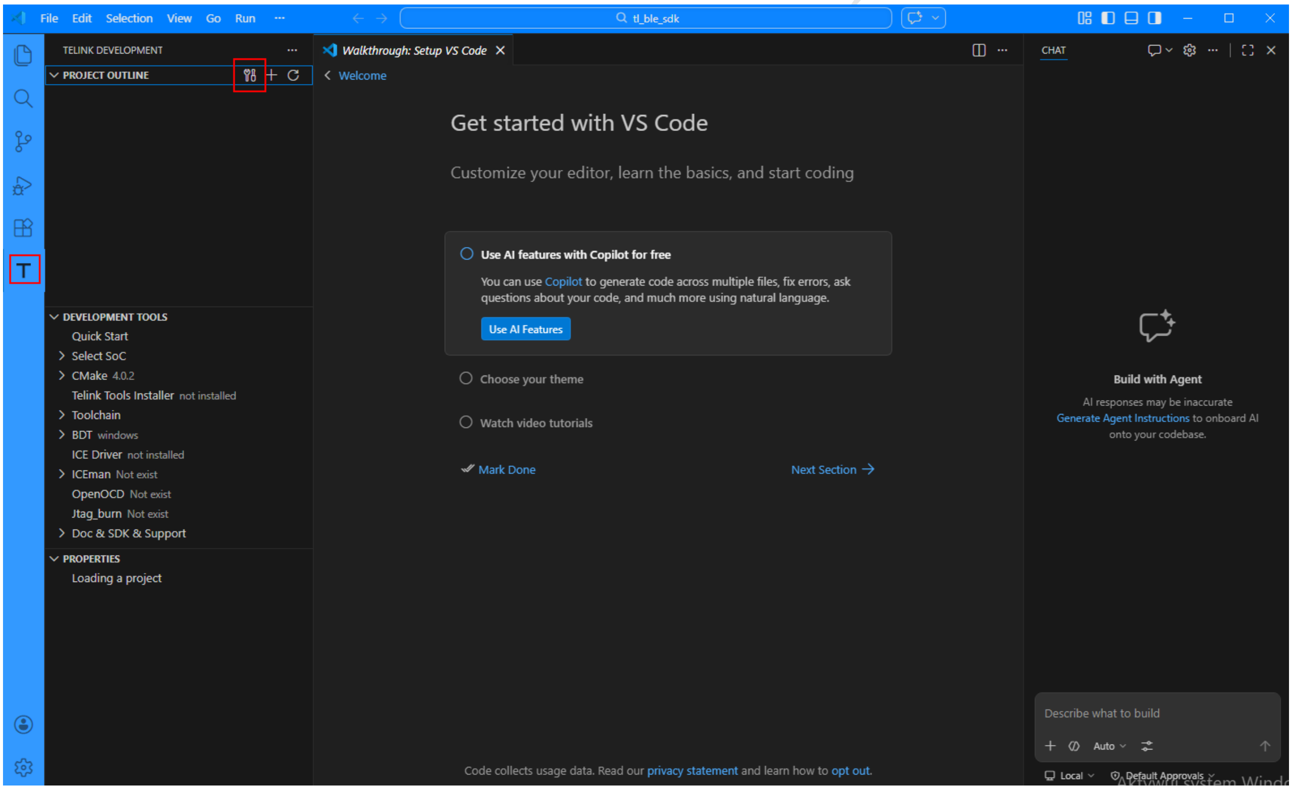

Figure 6.10: outline_and_convert

Select proper SoC (TL323X in this case)

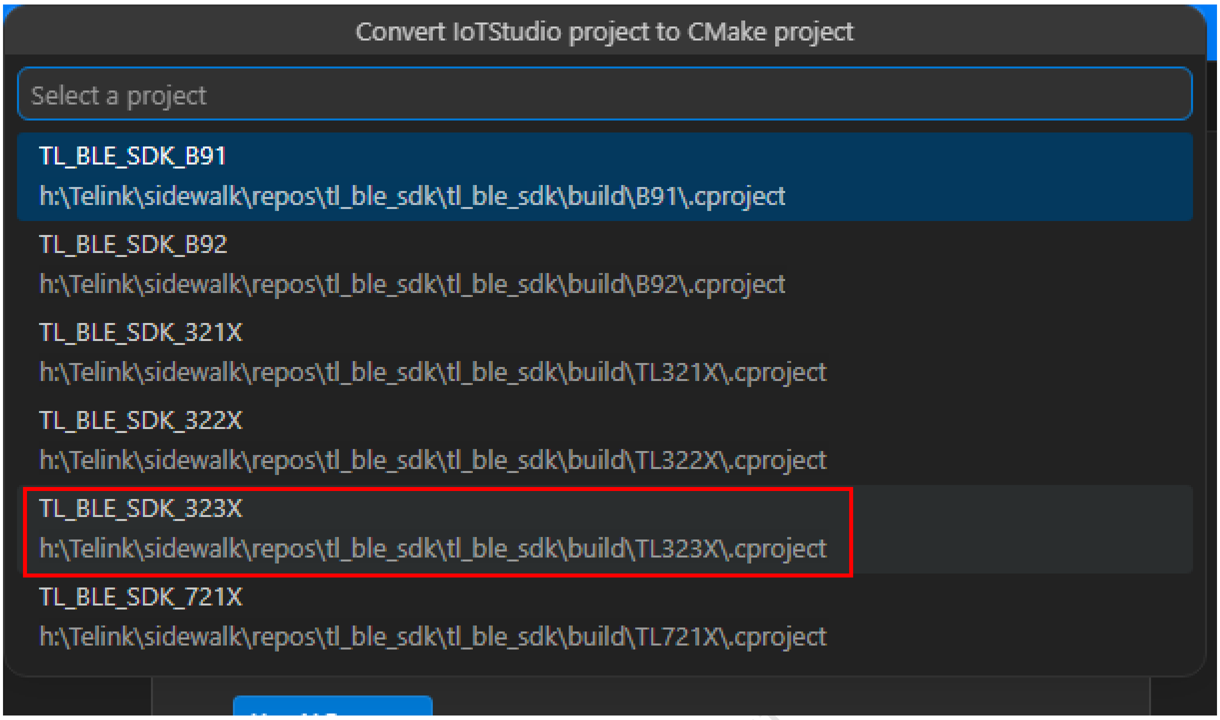

Figure 6.11: tl323x_selection

Build desired build variant (Amazon_sid_dut for example) by clicking rocket button next to build variant name.

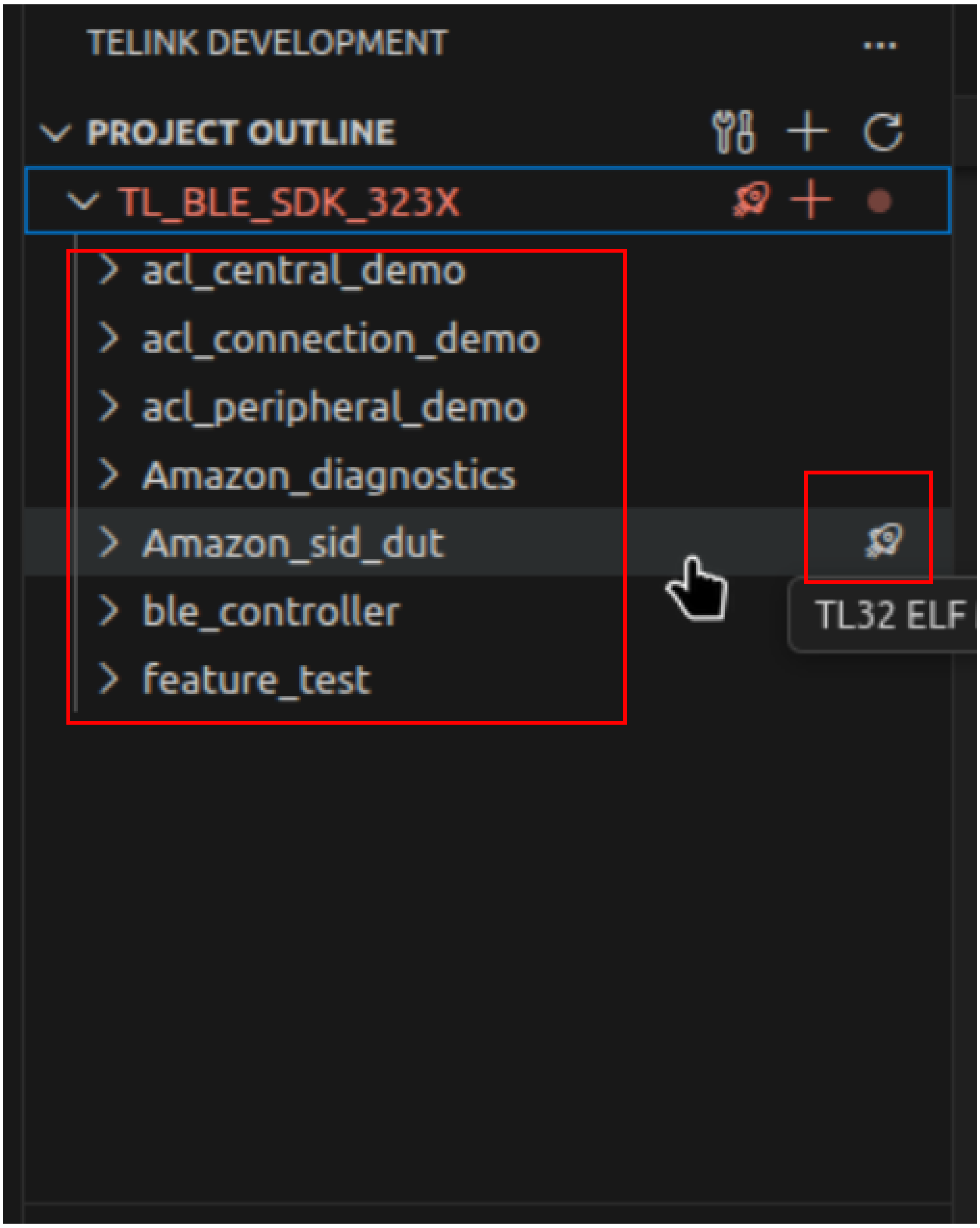

Figure 6.12: start_build

Build process should start and build artifacts should be visible on final.

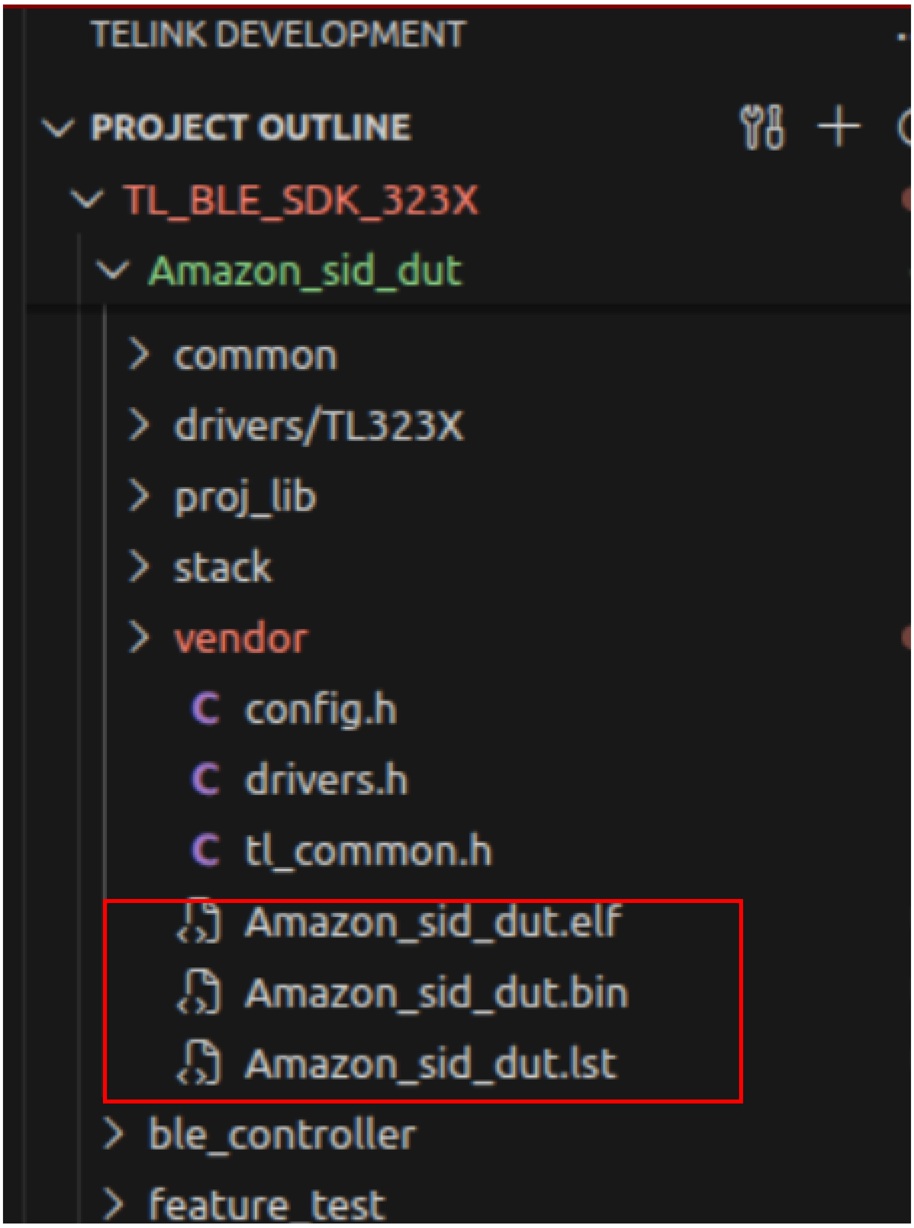

Figure 6.13: build_artifacts

6.6 Flash the Board

6.6.1 Address Map

Memory layout for the TL323X (ML3238S module) with 1024 KB flash.

Address	Size	Contents	Notes
`0x00000000`	~412 KB max	Firmware binary	Varies by demo (see below)
`0x000F5000`	—	(Legacy MFG address)	**Do not use** — changed in V1.0.0.6

Address	Size	Contents	Notes
`0x000F9000`	~4 KB	MFG provisioning blob	Device certificates and keys

Note:

- **Breaking change in V1.0.0.6**: The MFG blob address moved from `0x000F5000` to `0x000F9000`. If upgrading from an earlier SDK version, update your flashing scripts accordingly.

6.6.2 Firmware Sizes by Demo

Measured at SDK V1.0.0.6:

Demo	Flash (bin)	IRAM	DRAM
Amazon_sid_dut	412 KB	111.3 KB	22.90 KB
Amazon_sid_sbdt	352 KB	109.42 KB	18.15 KB
Amazon_diagnostics	165 KB	74.1 KB	12.75 KB

6.6.3 Flashing Procedure

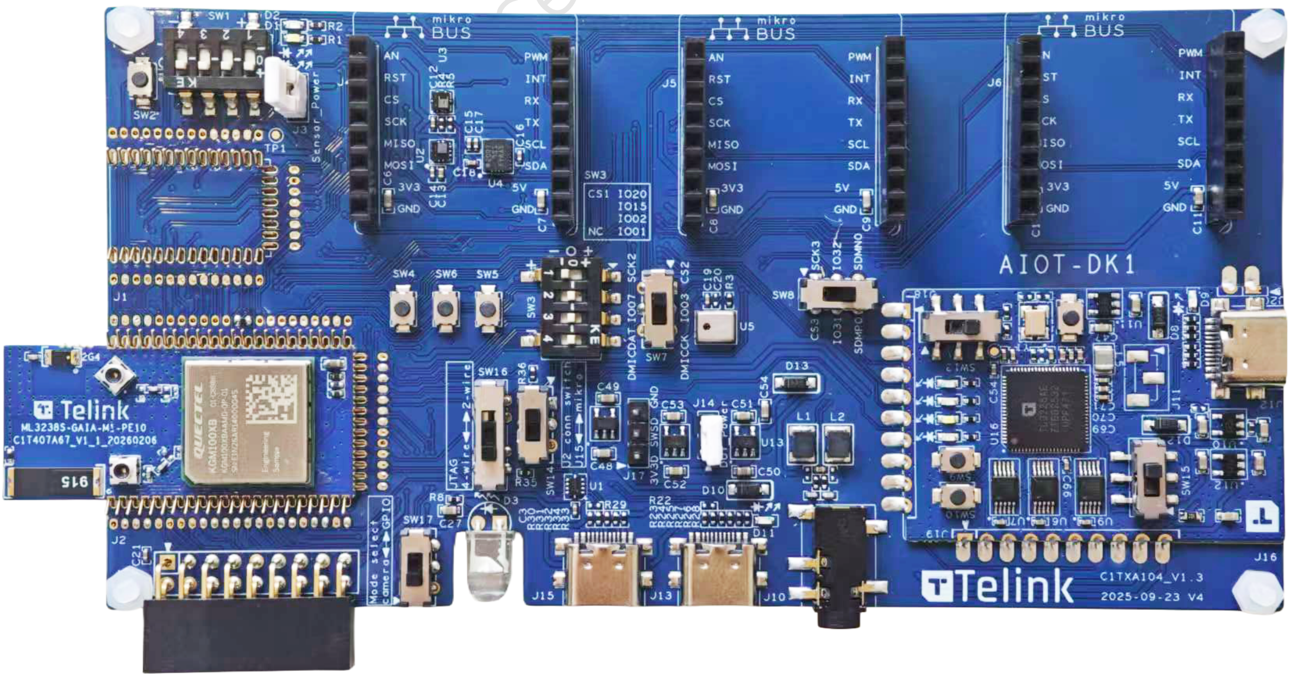

Figure 6.14: AIOT-DK1 board with ML3238S module

Attach ML3238S module into AIOT-DK1 board. Pay special attention for pin alignment. We have two main programming methods: one using the Telink programming tool, and the other using the AIOT-DK1 On-board Programmer.

The following demonstration uses the Telink programming tool. For more programming details, please refer to link: Burning and Debugging Tool User Guide.

Connect BDT flashing tool to the board using J17 connector.

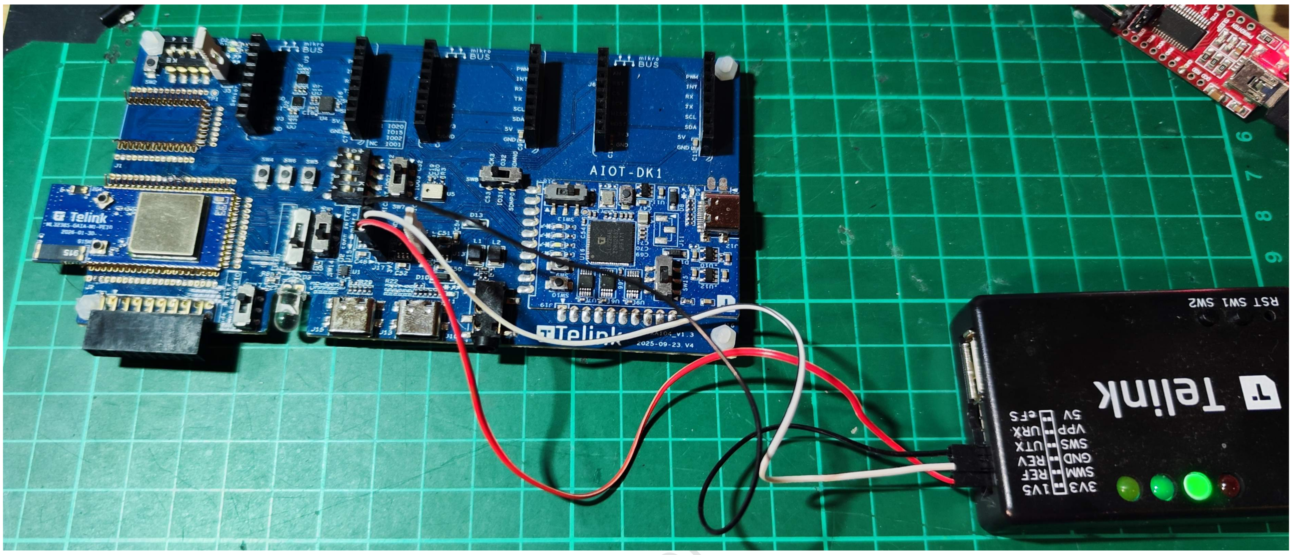

Figure 6.15: Flashing setup

In the picture above:

- RED - 3,3V
- White - SWS/SWM
- Black - ground

Note:

> - Use "Activate" function of BDT too, when flashing/erasing is not starting.

For out of the box experience, an application based on Amazon Sensor Monitoring has been prepared.

For now it is only available as a binary, which can be downloaded at sidewalk_sensor_monitoring.bin.

Click on BDT tool (found in Development Tools VS Code sidebar) and select proper BDT device you have.

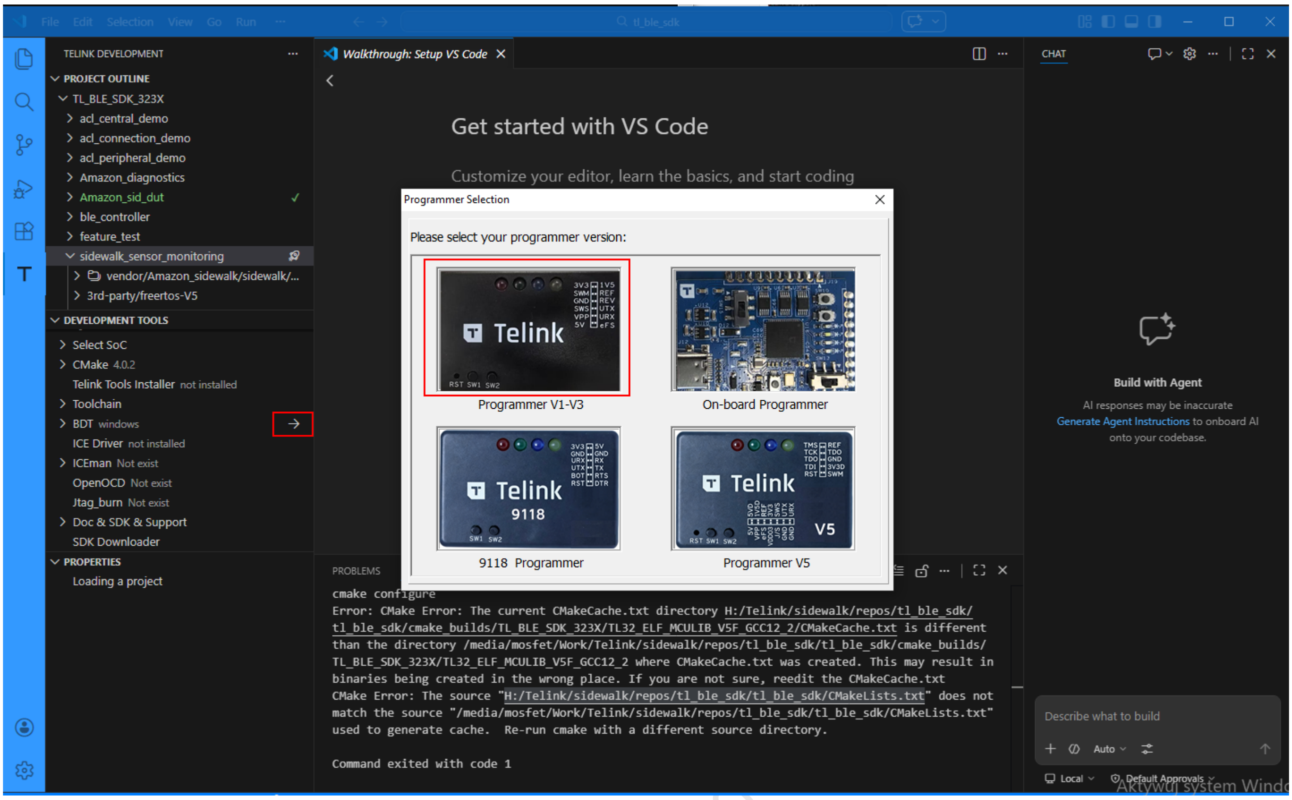

Figure 6.16: Start BDT

Make sure proper SoC (TL323X) is selected:

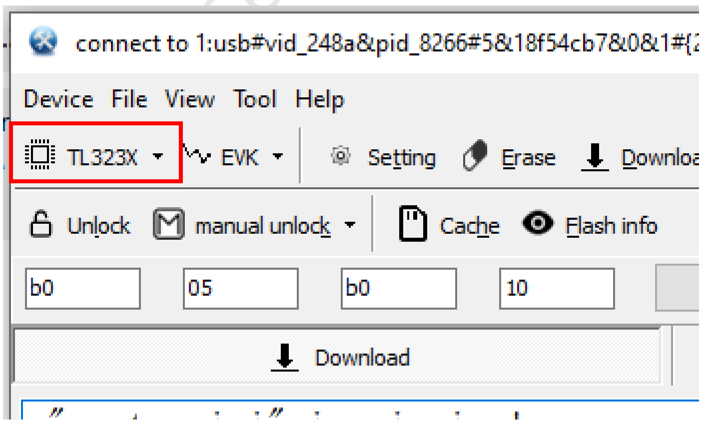

Figure 6.17: SoC Selection

Flash firmware binary alongside with MFG blob.

Firmware binary shall be put into 0x00000000 address.

The MFG blob shall be put into 0x000f9000 address.

To do this in comfortable way, BDT tool offers multi address download.

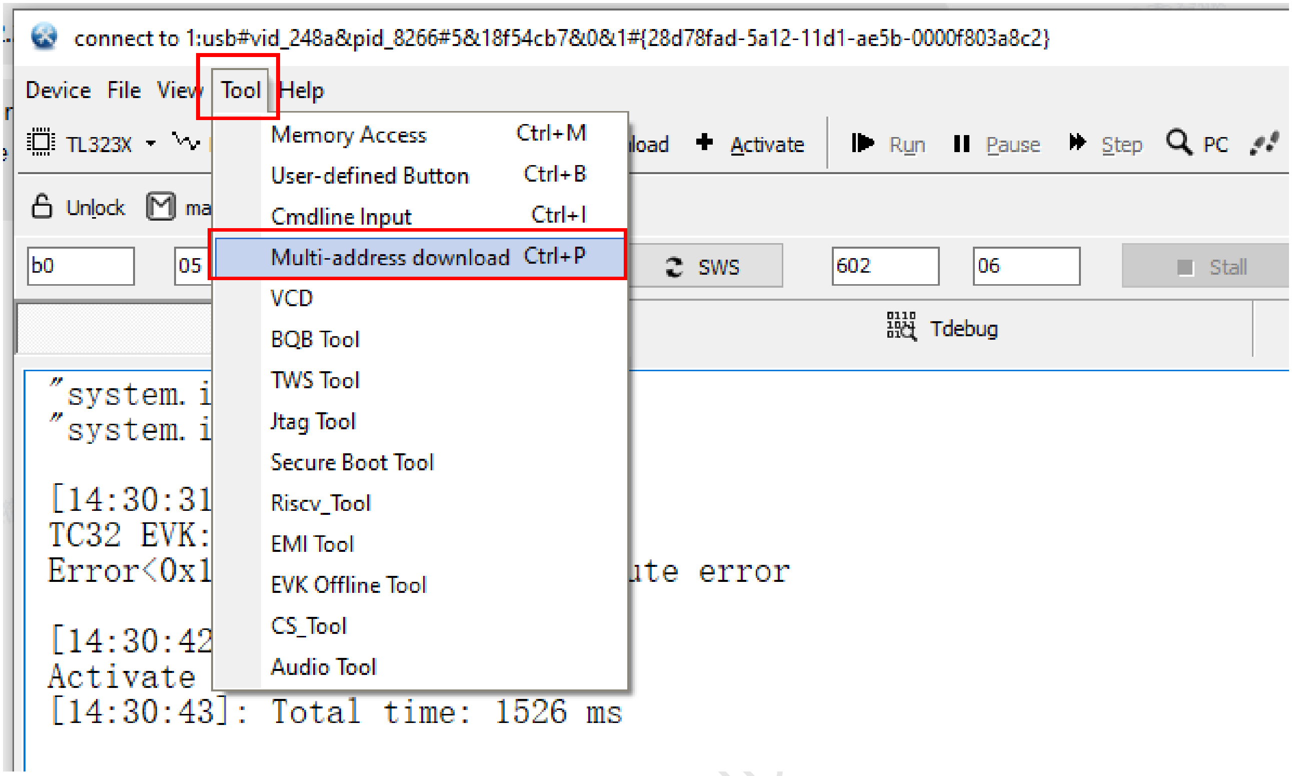

Figure 6.18: Multi Address

Provide paths to the binary and MFG blob and flash the board.

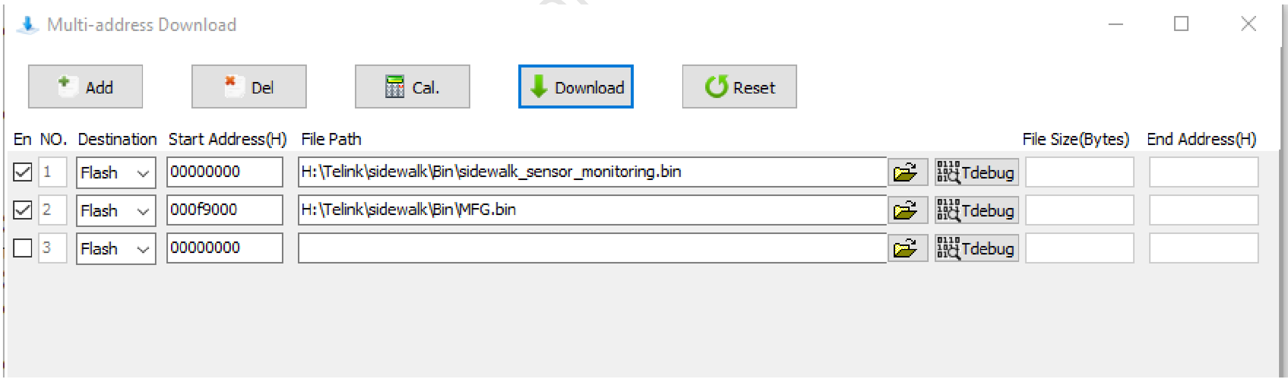

Figure 6.19: Multi Binaries

Hit “DOWNLOAD” button and wait until flashing is over.

For debugging purposes, UART->USB converter can be connected to PA4 (available at “RX” pin J4 “microbus” connector).

Logs are sent at 1Mbps baudrate.

Note:

- When uploading new MFG it is highly recommended to clean whole Flash memory of Telink SoC.

Make sure proper memory size is configured (1024k in case of ML3238S)

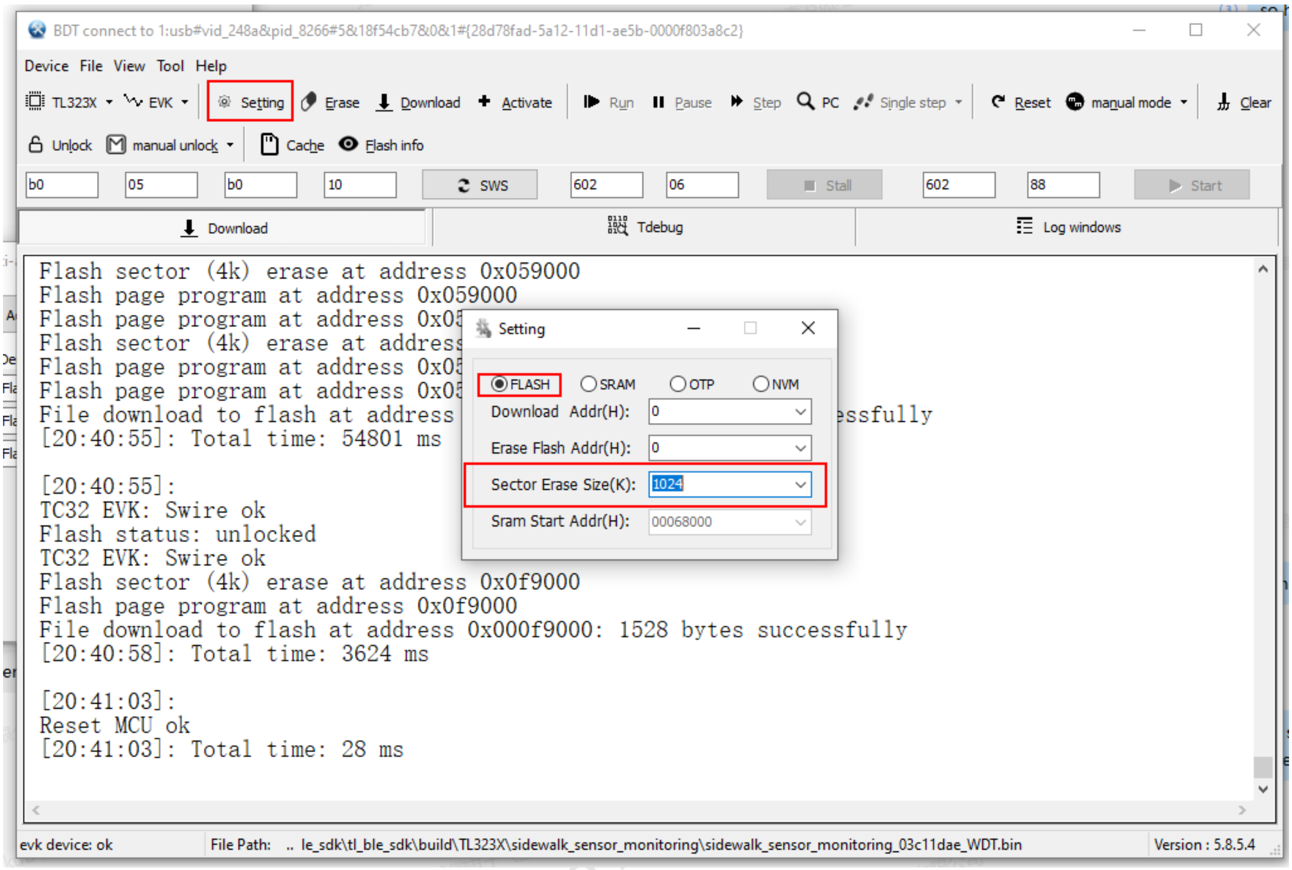

Figure 6.20: Flash Settings

Click “Erase” and wait until whole memory will be cleaned.

Now you can flash using multi address download tool:

File	Address
Firmware binary (e.g., `Amazon_sid_dut.bin`)	`0x00000000`
MFG blob (`mfg.bin`)	`0x000F9000`

Click **DOWNLOAD** and wait for completion.

6.6.4 OTA Considerations

- When the firmware binary exceeds **256 KB**, you must change the OTA startup address using:

```
blc_ota_setFirmwareSizeAndBootAddress(size, address);
```

This API call must be placed **before** `sys_init()` .

- There is a small probability of OTA failure in the current version (known issue).

6.6.5 Flash Protection

Flash protection is **enabled by default** in the SDK.

- During development: use the BDT **Unlock** command to access flash
- For mass production: you **must** enable flash protection in your application
- The SDK sample code provides a reference implementation — adapt it to your actual firmware size, OTA design, and data storage requirements

6.7 Test the Sidewalk Demo

Use J13 USB Type-C connector to power AIOT-DK1 board. Use SW2 button (reset) to start the firmware execution.

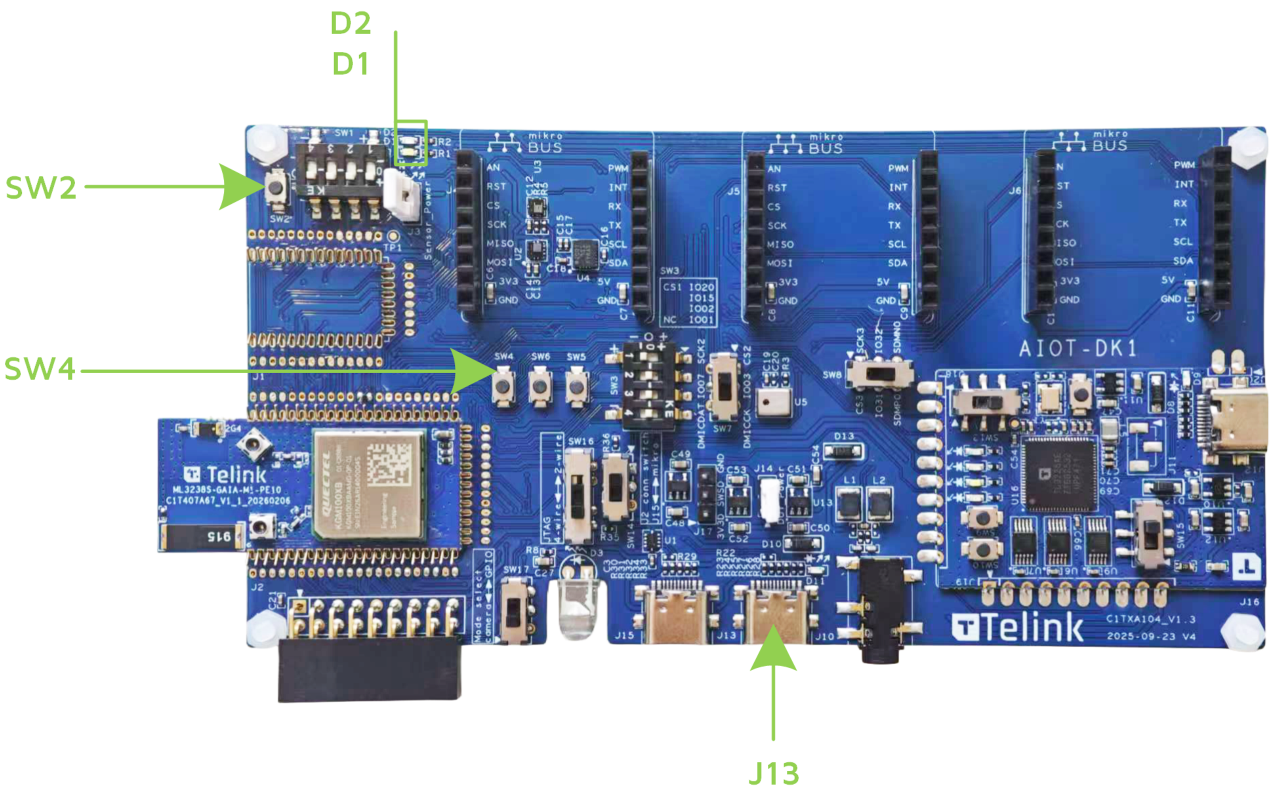

Figure 6.21: Test the Sidewalk Demo

The D1 and D2 LEDs should light up and go dark when sidewalk connection is established. After a while the web application (link gathered in Setup Amazon Cloud section) will notice Device appear, temperature reports shall be visible, SW4 button shall control "Push Button Device" and LED1/LED2 shall control white/green LEDs on the board.

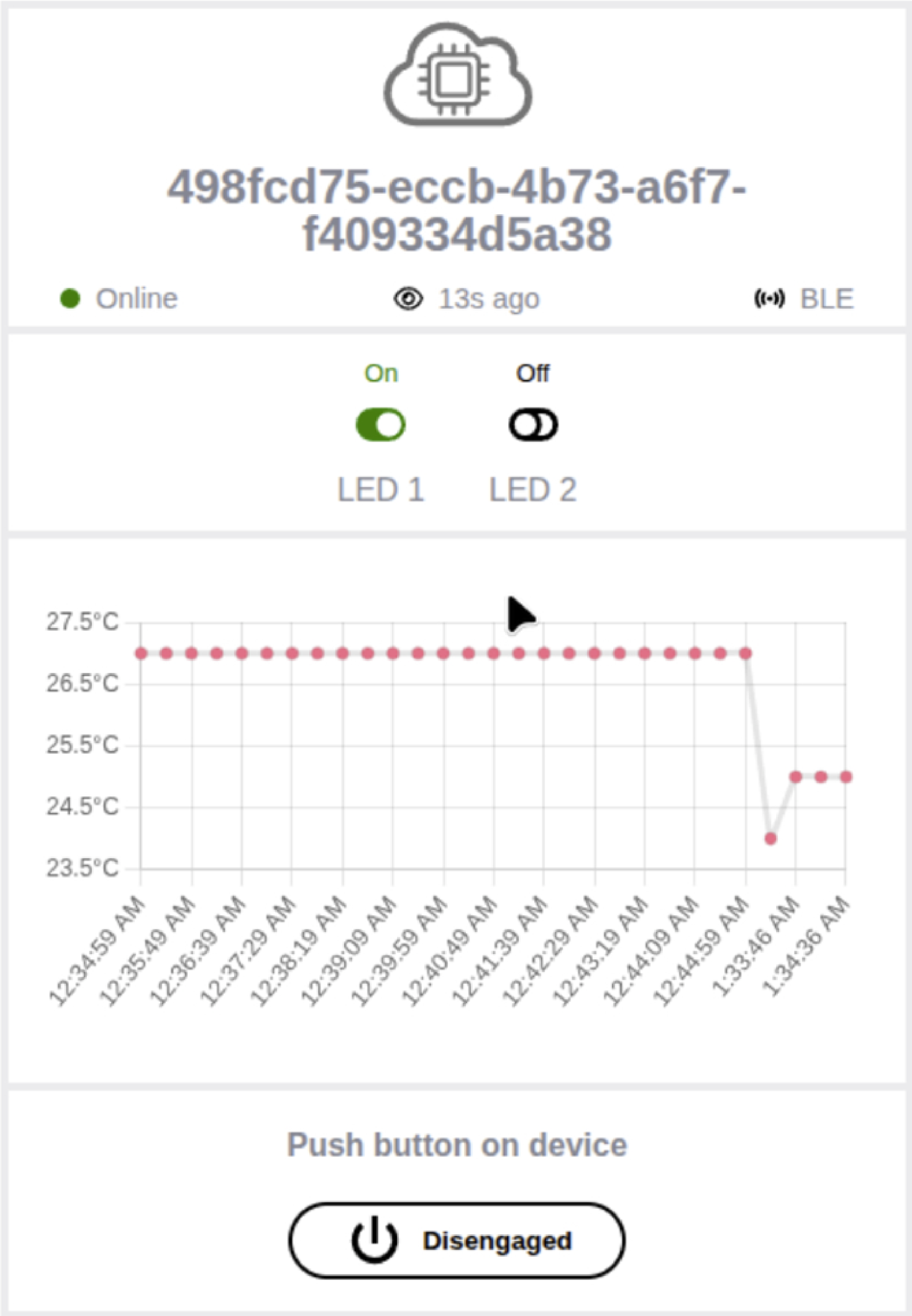

Figure 6.22: Web App

6.8 Test the DUT Demo

The DUT (Device Under Test) demo provides an interactive UART CLI for Sidewalk operations. This is the primary demo for development and testing.

6.8.1 Use Previously Build Demo

```
Amazon_sid_dut.bin
```

Refer to Build the Firmware.

6.8.2 Flash

Amazon_sid_dut.bin at 0x00000000

MFG.bin (used in previous demo) at 0x000f9000

6.8.3 Connect UART

Connect UART - USB converter connect to the AIOT-DK1 board as follows:

GPIO	Function	AIOT-DK1 location	USB Dongle Pin
PA1	TX (board talks)	J4 AN	RX (dongle listens)
PA2	RX (board listens)	J4 TX	TX (dongle talks)
GND	Signal ground	J4 GND	GND (dongle ground)

6.8.4 Open Terminal

Use terminal software of choice (like minicom or realterm etc.)

Configure on /dev/ttyUSBx or COMx: related to UART-USB dongle connected.

Use 115200 industry standard baudrate.

6.8.5 Reset the Board

You should see on the terminal:

```
[000b]sid QA cli application started...
[000c]starting asd_rtos_cli loop
-->
```

6.8.6 Connect to Sidewalk Network

Write a command into the terminal in order to initiate sidewalk stack in BLE mode:

```
sid init 1
```

After successful response:

```
[001d]CMD: ERR: 0
```

Write a command to register sidewalk stack:

```
sid start
```

Wait until successful time synchronization, it should look like:

```
[0077]Updated time: 1457484450.656250
```

6.8.7 Open the AWS IAM Console

Open your AWS IAM console (Amazon account) and navigate to AWS IoT -> MQTT test client. Use "#" wildcard to subscribe to all MQTT topics.

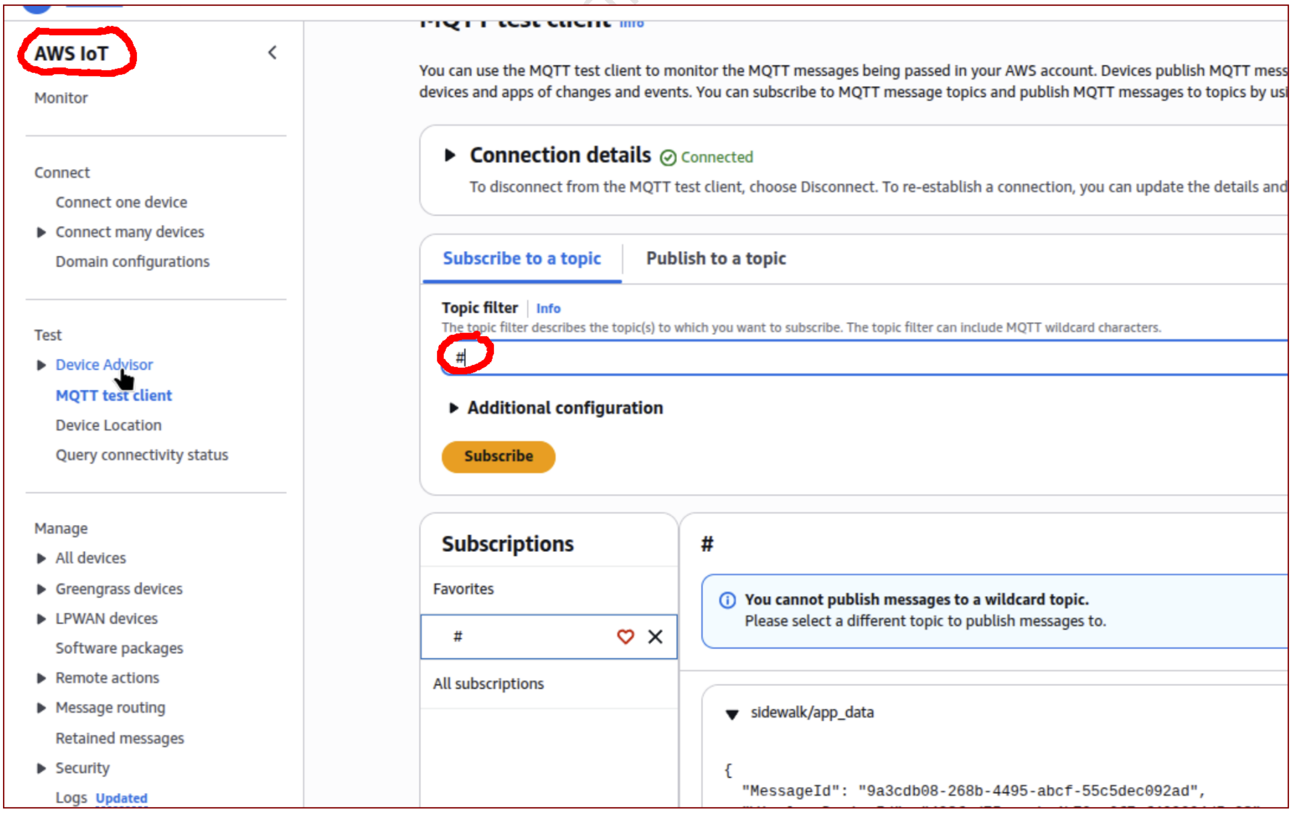

Figure 6.23: MQTT Test Client

6.8.8 Send Test Message

Using the UART console connected to Telink demo write a command:

```
sid conn_req 1
```

This should establish BLE connection between Telink Demo and the Gateway.

Wait until the log informing that connection is established:

```
[0158]EVENT SID STATUS LINK MODE: LORA: 0, FSK: 0, BLE: 1
```

Issue another command:

```
sid send -a 1 1 30 Hello
```

Go to MQTT test client on AWS IAM console:

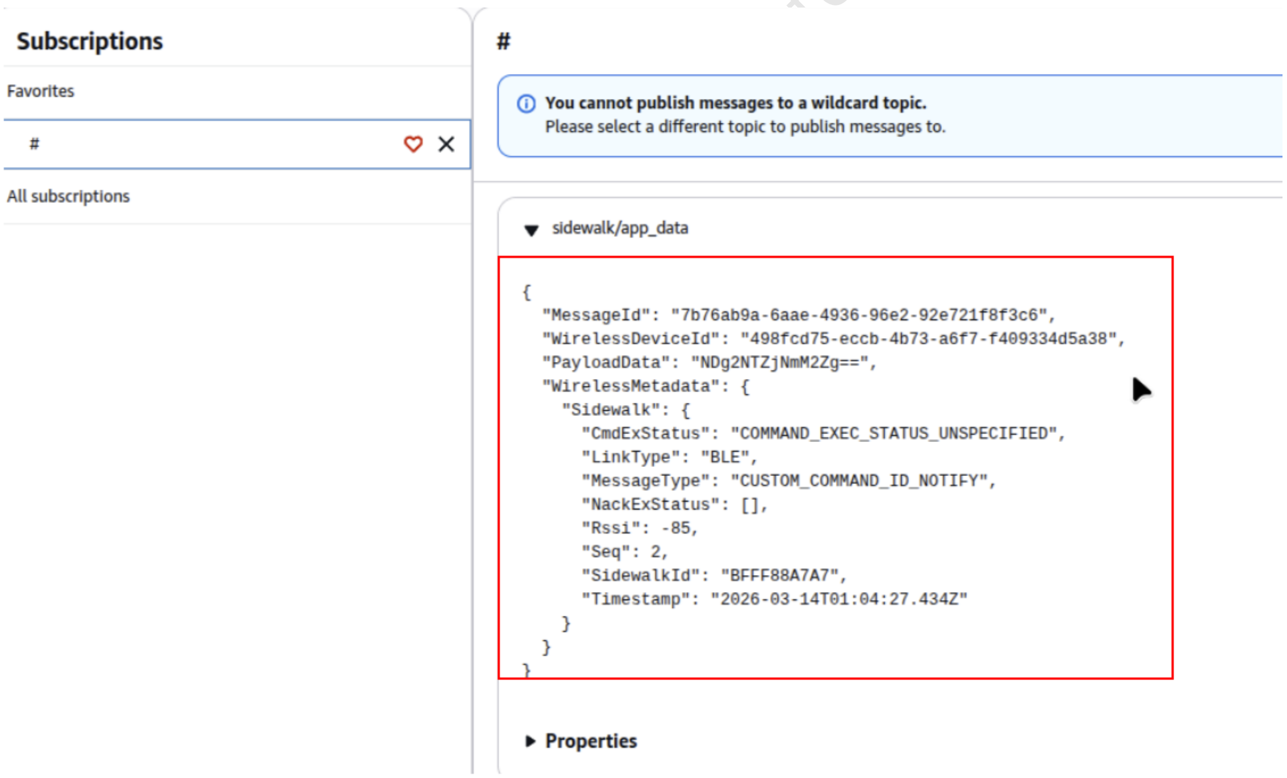

Figure 6.24: MQTT Test Client

```
NDg2NTZjNmM2Zg==
```

You can use widely available Base64 to hex converts to get:

```
34383635366336633666
```

after converting this hex to string:

```
48656c6c6f
```

And converting to string again:

```
Hello
```

6.8.9 Receive Test Message

Connect the board to sidewalk network like in Send Test Message from AWS CLI tool (used in web app deployment and provisioning steps) issue a command:

```
aws --region us-east-1 iotwireless send-data-to-wireless-device --id device_id_from_provisioning
--transmit-mode 0 --payload-data "SGVsbG8gVGVsaW5rIQ==" --wireless-metadata "Sidewalk={Seq=42}"
```

Observe UART console of Telink demo board. There should be a log like:

```
[00f0]Data: 48656c6c6f2054656c696e6b21
```

which can be converted to string.

Note:

- When instead of message data you find log line looking like this:

```
[02ac]msg dup: 4101000000 (42)
```

It means that message number 42 was already sent. All you need is to increase the number of "Sidewalk={Seq=xx}"

6.9 Troubleshooting

Common issues and solutions when working with the Telink Sidewalk SDK.

6.9.1 Flashing Issues

6.9.1.1 BDT won't Start Flashing or Erasing

Symptom: Clicking Download or Erase in BDT does nothing.

Solution: Use the BDT **Activate** function to wake the MCU, then retry.

6.9.1.2 Board not Detected During Flashing

Checklist:

- The board must **not** be connected to USB during flashing — power comes through J17 only
- Verify J17 wiring: Red = 3.3V, White = SWS/SWM, Black = GND
- Confirm the correct SoC (TL323X) is selected in BDT
- Set flash memory size to **1024K** for ML3238S

6.9.1.3 MFG Blob not Taking Effect After Reflash

Always **erase the full flash** before writing a new MFG blob. Residual data from a previous MFG can cause provisioning failures.

6.9.1.4 Wrong MFG Address

The MFG blob address changed in SDK V1.0.0.6:

- **Old address**: `0x000F5000`
- **New address**: `0x000F9000`

If upgrading from an earlier SDK version, update your flashing configuration.

6.9.2 VPN Setup for Non-US Development

Amazon Sidewalk is currently available only in the US, Canada, and Mexico. Developers in other regions must route their gateway's internet connection through a US-based VPN.

6.9.2.1 Proven Setup

- **Router**: GL.Inet GL-MT300N-V2 (Mango)
- **VPN provider**: NordVPN
- **Protocol**: OpenVPN

6.9.2.2 Steps

- Set up the GL.Inet router with NordVPN using the OpenVPN protocol
- Connect your Echo 4 gateway to the GL.Inet router's Wi-Fi
- The gateway will appear as US-located to Amazon's servers

Note:

> - No other VPN provider or protocol has been officially tested. Amazon recommends the OpenVPN protocol.

6.9.2.3 Sub-GHz in Europe

Sidewalk uses 902–928 MHz for sub-giga (FSK, LoRa). In the EU, this frequency range conflicts with LTE bands protected by regulatory agencies.

For sub-GHz development outside the US:

- Use an RF shielded cage or room
- Alternatively, convert the Echo 4 for cable operation using a u.FL SMA pigtail with a 40–60 dB attenuator (requires RF/hardware skills, not officially recommended)

6.9.3 AWS Cloud Issues

6.9.3.1 Deployment Fails with Permission Errors

The `generate_policy.py` script does not generate all required permissions. You need to add these additional inline policies to your IAM user:

- **S3 Bucket Tagging** — `s3:PutBucketTagging` , `s3:GetBucketTagging` , `s3:DeleteBucketTagging`
- **IoT Wireless Tagging** — `iotwireless:TagResource` , `iotwireless:UntagResource` , `iotwireless:ListTagsForRes`
- **IoT Wireless Full Access** — `iotwireless:\*`
- **AmazonDynamoDBFullAccess** — attach the AWS managed policy

See Set Up IAM Permissions for the full JSON policies.

6.9.3.2 MFG Blob from Provisioning Script is Incompatible

The MFG blob generated by `provision_sidewalk_end_device.py` is not directly compatible with the Telink demo due to format differences.

Solution: Use the Telink-specific provisioning script instead:

```
cd tl_ble_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/tools/provision/
python3 provision.py nordic aws --output_bin mfg.bin --certificate_json certificate.json --addr
0x1F4000
```

6.9.4 Sidewalk Connection Issues

6.9.4.1 Device doesn't Connect to the Gateway

Checklist:

- MFG blob is correctly flashed at `0x000F9000`
- Gateway (Echo 4) is online and registered as a Sidewalk gateway
- Gateway and device are within BLE range (~30 ft / 10 m for reliable connection)
- If outside the US: VPN is active on the gateway's network
- Full flash erase was performed before writing the MFG blob

6.9.4.2 Time Sync Never Completes

After `sid start`, the device must synchronize time with the Sidewalk network. If `Updated time:` never appears: - Verify the gateway is online and connected to the internet - Move the device closer to the gateway - Check that the MFG blob matches the device profile in AWS IoT

6.9.4.3 `msg dup` when Receiving Downlink Messages

```
[02ac]msg dup: 4101000000 (42)
```

This means the sequence number has already been used. Each `Seq` value in the `send-data-to-wireless-device` command must be unique. Increment the number:

```
--wireless-metadata "Sidewalk={Seq=43}"
```

6.9.5 Build Issues

6.9.5.1 Projects not Visible in VS Code Extension

After opening the SDK in VS Code, if the Telink extension's Project Outline is empty: - Click the **Convert** button in the Project Outline sidebar - Wait a few seconds for projects to populate - Select **TL323X** as the target SoC

6.9.5.2 CMake or Toolchain not Found

When clicking the build (rocket) icon: - If prompted "No find toolchain": click **Configure** and point to your Telink IoT Studio toolchain path (e.g., `.../toolchains/nds32le-elf-mculib-v5f/bin/riscv32-elf-gcc`) - If prompted "CMake is not available": click **Install** and wait for completion (~1 min)

6.9.6 UART / Debug Log Issues

6.9.6.1 No Output on UART Console

- **Sensor monitoring demo**: uses PA4 at **1,000,000** baud (1 Mbps) — most terminal apps default to 115200
- **DUT demo**: uses PA1 (TX) / PA2 (RX) at **115,200** baud
- Verify the correct GPIO pins and baud rate for your demo
- Check that TX/RX are not swapped between the board and USB dongle

6.9.6.2 Garbled Output

Wrong baud rate. Double-check: - Sensor monitoring: 1,000,000 - DUT: 115,200

6.9.7 Known Issues (V1.0.0.6)

- ACL peripheral connection may fail with low probability
- OTA has a small probability of failure
- SBDT demo: abnormal operations during upgrade may cause system exceptions
- Firmware >256 KB requires calling `blc_ota_setFirmwareSizeAndBootAddress()` before `sys_init()`

7 Complete Sidewalk Demonstration

This chapter primarily introduces a demonstration solution for Amazon Sidewalk functionality based on the Telink AIOT-DK1+ML3238S development kit. It details the implementation process for core features, including device-mobile app connectivity, remote control, sensor data collection, and real-time visualization.

7.1 Hardware and Software Preparation

Figure 7.1: Equipment Required for the Demonstration

(1) Wi-Fi Router

- The Sidewalk Gateway must connect to the network via Wi-Fi to transmit data to the AWS Cloud.
- Access Restrictions: As of March 2026, direct access is restricted to the United States, Canada, and Mexico. Users in other regions are required to connect to the server using a VPN.
- This example uses the Spitz AX (GL X3000). It is a Wi Fi 6 cellular gateway that supports VPN connections and dual SIM card slots, providing reliable internet connectivity for Sidewalk Gateway deployment.

(2) Sidewalk Gateway

- Core Function: Enables data transmission between Sidewalk end devices and the AWS Cloud.
- For the detailed gateway information, refer to the List of Sidewalk Gateways.

(3) AIOT-DK1 + ML3238S Development Kit

- This demonstration uses the AIOT-DK1 as the standard development baseboard. The ML3238S development daughterboard is directly compatible with the baseboard's J2 socket.
  - AIOT-DK1 User Manual
  - ML3238S User Manual (TBD)
- Test-related bin files to be burned (Currently for Telink internal use only)
  - Bin file for burning the test firmware, address 0x00: sidewalk_sensor_monitoring.bin
  - Bin file for burning the MFG configuration file (each device requires a dedicated MFG file generated separately), address 0xF9000 (1MB Flash): Sidewalk_mfg.bin

Note:

> - Please refer to the Appendix: Sidewalk SDK Overview for firmware generation instructions and replace them with your own test firmware.

(4) Mobile app

- Compatibility Requirements: Supports only iPhone devices running **iOS 15 or later**.
- Download and Installation: TelinkSidewalk iOS (iPhone/iPad) Version IPA Download - PGYER.com

7.2 Operation Process

The data transmission process is shown below:

Figure 7.2: Data Transmission Process

7.2.1 Account Registration and Sign In

Users should complete account registration and sign in manually when using the TelinkSidewalk app for the first time. The app remains signed in by default on subsequent openings.

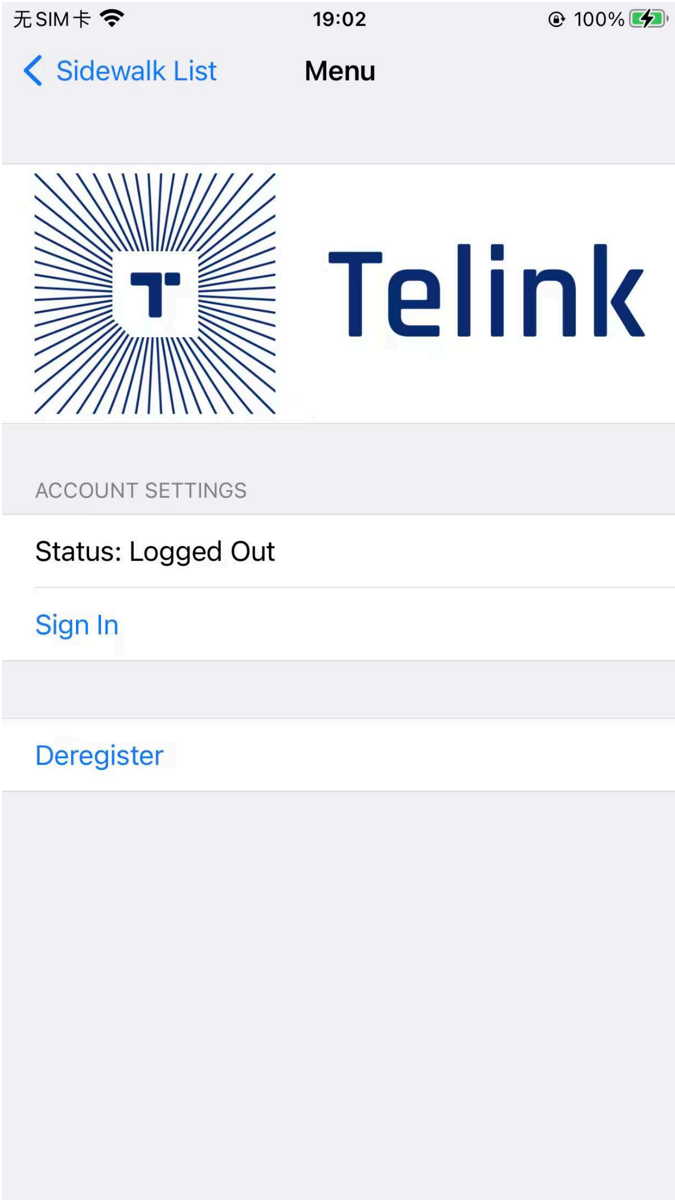

Figure 7.3: TelinkSidewalk App Menu Interface

Note:

- The same account supports simultaneous sign-in across multiple Apple devices without affecting device connections or operations.

7.2.2 Device Registration and Deregistration Process

When connecting a device to the gateway for the first time, complete the device registration process. After registration, the device defaults to the `Registered` state and does not require re-registration for subsequent connections.

- Registration requires enabling the phone's BLE function and scanning the device's Bluetooth advertising. Locate the device advertising the name "SID_APP", select it, and then click `Register` to complete the process. The device will be registered as an LPWAN device.

- After registration, the device is added to the device list, and users can find the corresponding device through this list.

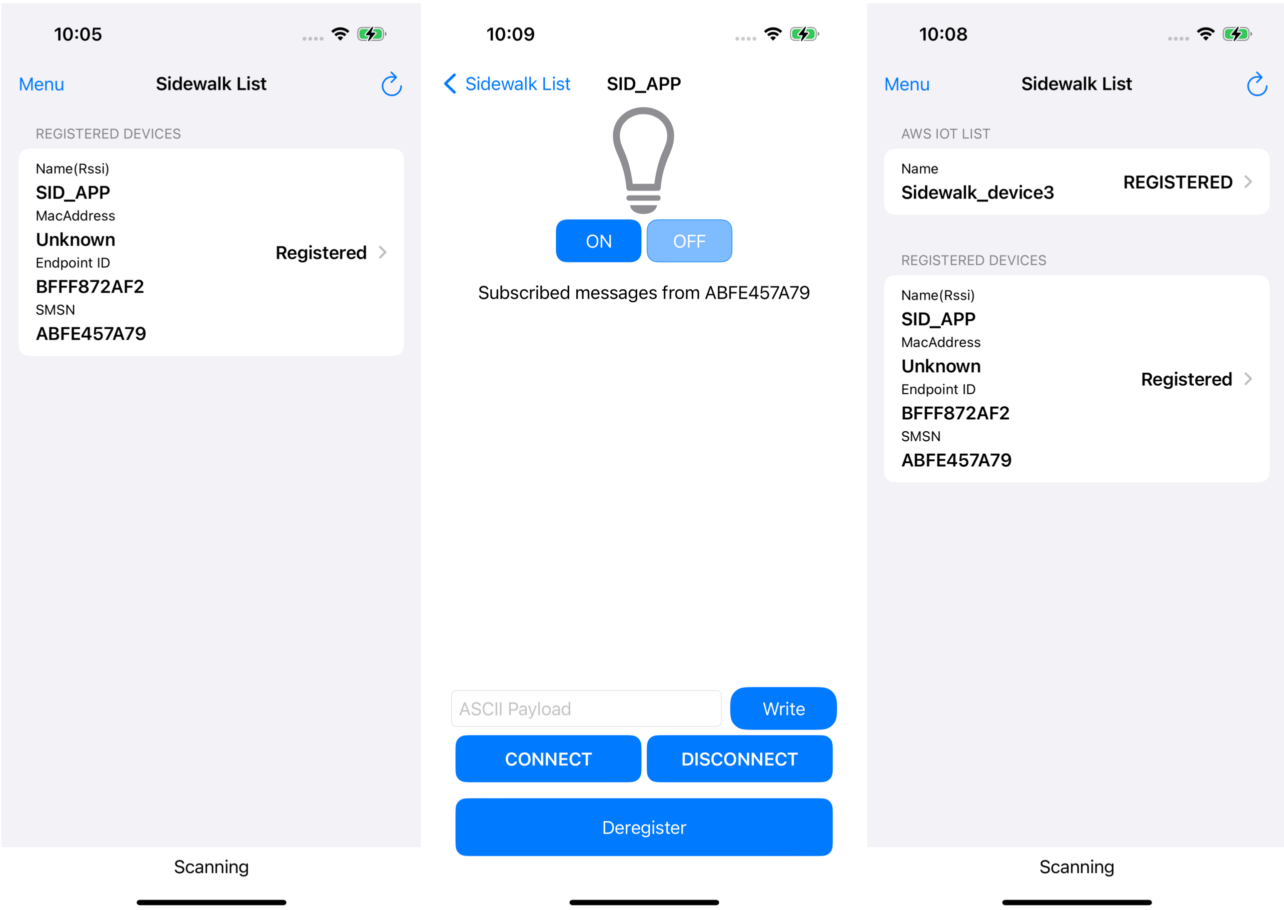

Figure 7.4: Device Registration Process

For deregistration, follow the process described above.

7.2.2.1 View Operation Logs

The app supports a `History Message` function for viewing historical device operations and event records.

7.2.2.2 Remote LED Control (Mobile App -> Device)

By clicking the LED0/LED1 control buttons on the app page, control commands are sent to the AWS cloud. These commands are then relayed via the gateway to the device. Through remote control, the LEDs on the AIOT-DK1 development board can be switched on or off.

7.2.2.3 Automatic Sensor Data Collection and Display (Device->Mobile App)

The device automatically uploads sensor data to the AWS cloud via the gateway. The app parses and displays this data in real time, including the following information:

- Temperature sensor data: Environmental temperature collected by the device (unit: °C), displayed in the app.
- IMU sensor data: Three-axis acceleration data (Acc X/Acc Y/Acc Z) displayed in the app.

7.2.2.4 Real-Time Button Status Feedback (Device->Mobile App)

When the SW4 button on the development board is pressed, the device sends the button trigger signal to the AWS cloud via the gateway. The mobile app retrieves this information from the cloud and displays it.

7.2.2.5 Supports BLE, SubG-FSK, and SubG-CSS Mode Switching

The working mode can be easily switched via the toggle button on the mobile APP.

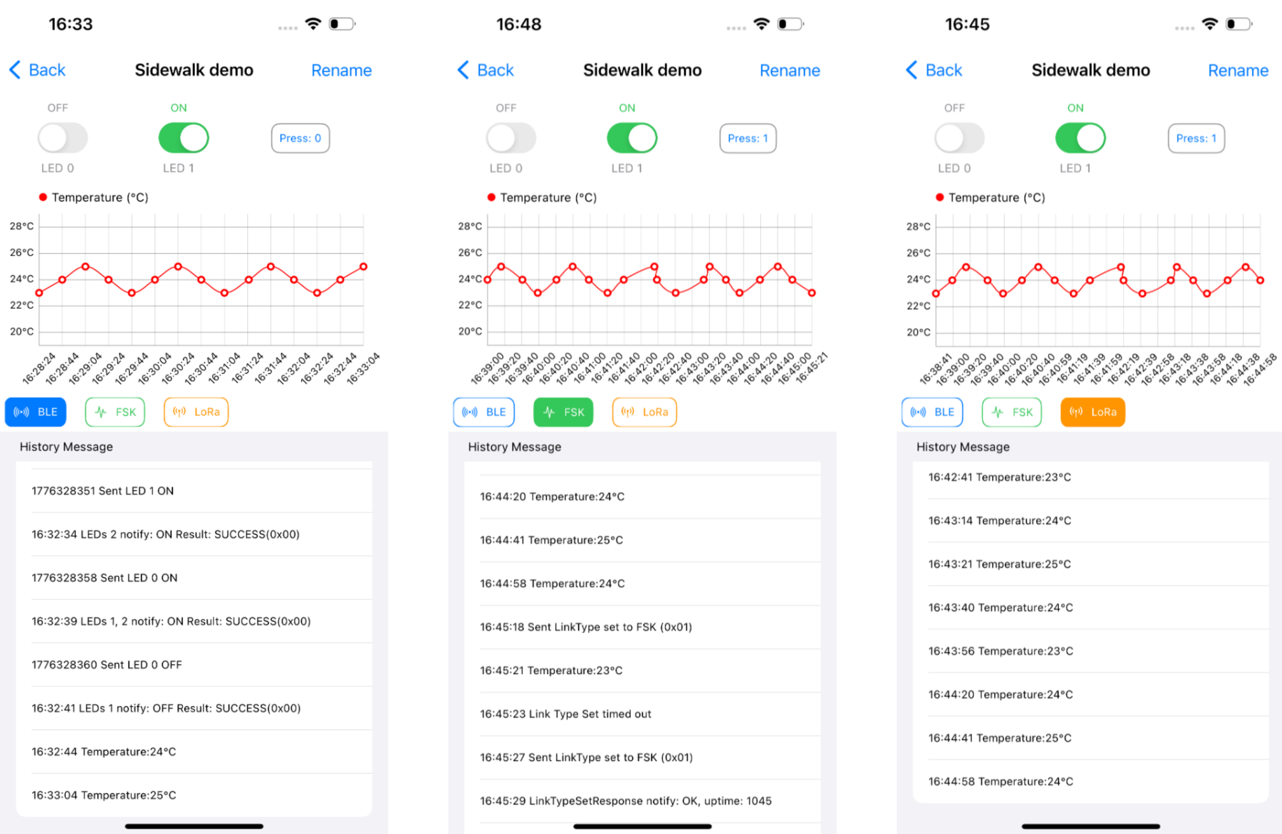

Figure 7.5: Device Data Interface

8 FAQ

8.1 Network Coverage & Availability

Q1: Where is Amazon Sidewalk available?

A1: Amazon Sidewalk is currently live in North America (US, Canada and Mexico).

Coverage maps are available for this region: https://coverage.sidewalk.amazon/.

European availability is planned for 2026, but has not yet been announced. Messaging for EU (European Union) audiences should avoid implying current coverage.

Q2: What is the expected range of a Sidewalk device?

A2: Range varies by radio protocol: BLE 10–50 m, SubG-FSK 100–250 m, SubG-CSS up to 2–5 km, depending on line of sight.

8.2 Product Compatibility & Interoperability

Q1: Can LoRaWAN end nodes support Sidewalk?

A1: Not natively. Sidewalk and LoRaWAN are separate protocols. Semtech chips support both protocols, allowing device manufacturers to build dual-protocol devices that operate on both LoRaWAN and Sidewalk networks when the appropriate firmware and certifications are implemented.

Q2: Can LoRaWAN gateways serve as Sidewalk endpoints?

A2: No. Sidewalk relies on Amazon gateway infrastructure, primarily Ring and Echo devices. LoRaWAN gateways operate as a separate infrastructure. Gateway-level interoperability between the two networks is not currently supported. For more details, see: https://docs.sidewalk.amazon/introduction/sidewalk-gateways.html.

Q3: How do I build a device that supports both LoRaWAN and Sidewalk?

A3: Semtech provides LoRa chips capable of supporting both protocols at the RF level. Developers should implement both protocol stacks in firmware and complete certification requirements for each network independently. Semtech and the Amazon Sidewalk team can provide technical guidance.

Q4: Are product SKUs hardware-identical across regions?

A4: Product SKUs may use identical hardware across regions but require region-specific firmware configuration. Automatic roaming across regions has not been confirmed. Developers should validate requirements with Semtech and the Sidewalk team.

Q5: Which Ring and Amazon devices act as Sidewalk gateways?

A5: Select Ring and Echo devices act as Sidewalk Bridges (gateways), extending the network. The specific models supporting this feature are listed in Amazon's Sidewalk documentation. See: https://docs.sidewalk.amazon/introduction/sidewalk-gateways.html.

8.3 Device Development & Certification

Q1: What does it take to design and certify a Sidewalk product?

A1: Use compatible hardware from a Sidewalk silicon partner (BLE/SubG-FSK/SubG-CSS-capable modules). All required tools, sample apps, and SDKs are available directly from silicon partners - there is no separate protocol stack to implement. Devices should complete Amazon's certification process. See silicon partner resources here.

Q2: What security models does Sidewalk support?

A2: Sidewalk uses a layered security architecture including device authentication, end to end encryption, and key management. Supported security mechanisms include certificate based identity and secure element integration. Full technical documentation is available through the Sidewalk developer portal.

Q3: What data rates, capacity, and latency can I expect?

A3: Performance depends on the selected radio mode. LoRa CSS delivers long range connectivity with low data rates and low power consumption, making it suitable for sensor telemetry. BLE provides higher throughput over shorter distances. Sidewalk is optimized for IoT workloads rather than real time or high bandwidth applications. PHY rates by radio protocol (note: actual throughput is lower in practice):

- BLE – 1 Mbps
- SubG-FSK – 50 kbps
- SubG-CSS (SF11) – ~2 kbps

8.4 Use Cases & Applications

Q1: What are the primary use cases for Sidewalk?

A1: Representative use cases include asset tracking, such as item or vehicle tracking, car alarm systems, smart home sensors, environmental monitoring, agriculture, industrial IoT, and smart city infrastructure.

Q2: Does Sidewalk support geolocation?

A2: Yes. Sidewalk supports BLE, Wi-Fi, and GNSS geolocation.

Q3: What AWS services does Sidewalk integrate with?

A3: Sidewalk integrates with AWS IoT Core. Device telemetry can flow into the broader AWS platform, including services such as AWS Lambda, Amazon S3, and Amazon DynamoDB. This enables developers to build scalable IoT applications on top of the Sidewalk network.